

Knowledge-Guided Adaptation of Pathology Foundation Models Effectively Improves Cross-domain Generalization and Demographic Fairness

Corresponding Author: Professor Lequan Yu

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #2

(Remarks to the Author)

This manuscript discusses the impacts of foundation models on computational pathology approaches, highlighting biases that are appearing in many of these models. Of note, there are out-of-domain data issues, where models overfit to a subset of the data on which they are trained, and do not generalize well to new tasks, which suggests that these foundation models are not learning the true underlying biology in the pathology slides. The authors perform a comprehensive analysis of some of these biases and proposed a method, FLEX, to correct for them. FLEX is applied to 16 predictive tasks in 4 different TCGA cancer data sets. FLEX shows statistically significant improvement in reducing non-clinically-relevant biases in pathology slide data, notably clinical site biases that are not relevant to the scientific questions the model is being trained to answer but which often confound AI and ML models. This improved method is compared to current standard approaches. The literature is well cited and the manuscript clearly written and easy to follow. Similarly, the figures are beautiful. The work supports the conclusions however additional evidence, outlined in my comments below, would make the claims stronger. As the manuscript is written now, it is unclear if these results will translate to other datasets or if they are specific to TCGA data. There are correlations between clinical sites and other demographic factors that may impact the results, creating relationships between the different sites discussed in the manuscript. The authors do not take these into account when stratifying the data to create IND and OOD subsets of the data. Overall, it is a well written manuscript addressing a significant concern in the field.

Comments:

The sites are not defined until the supplemental text. Please make this clearer earlier in the text, that sites reference the literal clinical sites that processing the samples.

Based on the results, it looks more like stain normalization works for some but not all sites - what site is out of the norm? Why? Please discuss.

On page 6, please reference the supplemental text and tables describing the 16 tasks.

The racial differences in performance are interesting. TCGA also has information about genetic ancestry (see <https://doi.org/10.1016/j.ccell.2020.04.012>). It would be interesting to see performance differences across genetic ancestry groups, as well as racial stratification. Doing this avoids misclassifying patients with ancestry from multiple human populations (common in African Americans in the TCGA data). This would be a better way to stratify the samples. There are also site-specific race and other demographic factor correlations. How is this handled in the analysis? Are training dataset sizes held constant?

Several of the biomarker prediction tasks (e.g. BRCA HER2, ER, and PR) have known racial biases. Others are not clearly known. Please discuss the variations in performance given known biases and performance improvements relative to the tasks - is there a larger performance gain in tasks where there are known racial biases?

It's unclear how much of the model performance is dependent on dataset biases beyond demographic data. Ideally, the trained model would be applied to non-TCGA and racial bias and other biases analyzed in this way. Other datasets exist with pathology slide images, and models trained in this manuscript could be applied to those.

Figure 2:

- Why is figure 2 referenced before figure 1?
- Figure 2 plots should have axis labels.
- In Figure 2, UMAP is a visualization technique that is affected by parameter tuning. How much of that is causing the biases here?
- How are the site stratifications actually done, as visualized in Fig 2b? This visualization doesn't explain it well. How were sites grouped? What about site-site relationships? For example, is there a relationship between the clinical equipment, the location, or something else?

Figure 3:

- Please move figure legends under the matching figures.
- It's unclear exactly what Figure 3a is plotting. Please provide more clarity in the caption description.
- Please add a horizontal line for the avg performance in the data overall, in Figure 3b.
- Plots used in Figure 3c are not appropriate for the data being modeled. Please update these to not use dynamite plots.
- For all of figure 3, please define the error bars and other data annotations for all subfigures.

Figure 4:

- Please do not use dynamite plots for Figure 4a. Boxplots, violin plots, or another visualization are a more representative way of visualizing these results.
- Please update the y-axis ranges to be consistent across each row in Figure 4a so it is easier to compare model performance across tasks.
- It's difficult to see the EBV +/- shapes in the Figure 4b plots. Please adjust to make these easier to distinguish.

Figure 5:

- Figure 5a missing a legend and the statistical tests is not defined. Are these adjusted for multiple hypotheses? Also, are the data repeatedly stratified then models tested? How much overlap is there in data between each test, and how is the significance of performance differences calculated? What size is the training data for each of these splits?
- Figure 5b would be more impactful if visualized differently, to show the change in performance across original, improvement, and FLEX. Perhaps using a bipartite graph.
- Additionally, there seems to be a large amount of variability in performance in Figure 5b, suggesting that there are other factors in the data splits driving performance differences. Please discuss this in the text.

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)

The authors developed a machine learning framework aimed at improving the generalizability of computational pathology diagnostic models. It focuses on the issue of site-specific tissue patterns and demographic profiles that hinder model performance across locations for data collection. Their approach uses visual and textual domain knowledge to guide an information bottleneck that filters out certain features and focuses on task-relevant ones. The framework is evaluated on binary classification tasks using data from The Cancer Genome Atlas (TCGA). Computational results show modest gains in the area under receiver operating characteristic curves (AUROC), particularly for performance in different participatory sites of the TCGA and reductions in performance gaps across demographic groups. Below are my comments.

General Comments:

The proposed method presents an interesting model design. There are a few potential issues regarding its scalability, interpretability, and practical utility. Notable limitations include reliance on human experts for patch selection, dependence on GPT-4o for concept generation without verification, and modest improvements in out-of-domain performance. In addition, the absence of statistical testing across tasks may undermine the method's robustness and its real-world applicability.

Specific Comments:

1. In the visual concept generation module, the selection of representative patches requires both GPT-4o filtering and manual validation by human experts. This dependency on expert input may limit the scalability and automation potential of the proposed method, particularly in clinical settings where expert time is costly and throughput demands are high.
2. The use of GPT-4o for generating textual concepts is not accompanied by a validation or correction mechanism. Given potential issues with hallucination and inconsistency in large language models, relying solely on GPT-4o outputs for guiding feature enhancement may introduce potential inaccuracies, especially in high-stakes medical applications.
3. Although the method is described as a cross-domain generalization method, the actual performance improvements on out-of-domain (OOD) datasets are modest. In some cases, the proposed method (Feature-Level Enhancement for Cross-domain generalization; FLEX) reduces the performance gap from only 4% to 3%, which may have limited clinical relevance or operational impact.
4. In Figure 2a, the UMAP visualizations show overlapping clusters even after applying the proposed method, indicating that site-specific signatures may not be fully eliminated. A quantitative measure of intra- and inter-class distances or silhouette scores would strengthen the claims of feature disentanglement and clustering quality.
5. Interestingly, standard stain normalization via the Reinhard method achieves better results than their methods in a few tasks. The authors could further discuss reasons leading to this discrepancy and address questions about the added value

of their method.

6. In Figure 2d and related visualizations, the authors mentioned their methods reduced attention paid to noisy regions. However, what constitutes noise was not clearly defined or objectively measured.

7. In Figure 3, the fairness gap (measured in AUROC differences across racial groups) ranges widely, sometimes as high as 0.7. Surprisingly, the proposed methods exhibit the widest performance spread in certain tasks. The authors could discuss the implications of this observation and the potential impact on the reliability of the proposed methods.

8. While the proposed method is compared against baseline and the standard Reinhard stain normalization method in Figure 3, no statistical tests (e.g., paired t-tests or Wilcoxon signed-rank tests) are reported to determine whether the observed performance differences are significant. The observed differences in results between methods appear minimal at times.

9. In Figure 4b, UMAP plots for the features obtained from the proposed methods still show substantial overlap between class labels across sites. The visual evidence does not compellingly support the claim of improved separability of diagnostic categories.

10. In Figure 5, performance improvements across different data scales and MIL model variants are relatively small, sometimes between 0.000 and 0.005 in AUROC. The authors could discuss the practical benefits of integrating the proposed methods into real-world pathology diagnosis.

(Remarks on code availability)

The code provides sufficient instructions for installing and running the application.

Version 1:

Reviewer comments:

Reviewer #2

(Remarks to the Author)

The additional analysis of two large datasets and subsequent results and manuscript updates largely address my concerns.

(Remarks on code availability)

The code for the initial submission is included but code relating to resubmission updates is missing from the repository. The code is well organized and the repository has a README with instructions to run the code.

Reviewer #3

(Remarks to the Author)

The authors have addressed the previously raised comments. Thank you.

(Remarks on code availability)

The authors provided a README file with instructions for installing and running the application.

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Responses to Reviewers' Comments for Manuscript NCOMMS-25-16314

**Knowledge-Guided Adaptation of Pathology
Foundation Models Effectively Improves Cross-domain
Generalization and Demographic Fairness**

Addressed Comments for Publication to
Nature Communications

Dear Reviewers,

We are pleased to resubmit a revised version of our manuscript entitled “Knowledge-Guided Adaptation of Pathology Foundation Models Effectively Improves Cross-domain Generalization and Demographic Fairness” (Manuscript ID: NCOMMS-25-16314) in response to the decision for **major revision**. We sincerely appreciate your efforts and the time dedicated to reviewing our paper. We have comprehensively addressed all comments and believe the manuscript has been substantially strengthened as a result.

To facilitate your review, we have adopted the following structure:

- The original comments from reviewers are presented in **blue boxes**.
- Our point-by-point responses follow each comment directly.
- Key sentences revised in the manuscript are highlighted within our responses in **gray boxes**.
- All changes within the revised manuscript file itself have been marked with **red text**.

The main revisions to the manuscript are summarized as follows:

- **External validation on two independent cohorts:** To address concerns about generalizability of our findings beyond the TCGA dataset (Reviewer #2, Reviewer #3), we have performed a new zero-shot evaluation on two large-scale, independent external cohorts: the public Clinical Proteomic Tumor Analysis Consortium collections (CPTAC; $n=1,738$) and our in-house dataset from Nanfang Hospital of Southern Medical University (NFH; $n=5,318$), totaling 7,056 new slides for external validation. The results (Fig. R6) demonstrate that *FLEX* achieves consistent and statistically significant performance improvements in real-world, out-of-distribution scenarios.
- **Expanded and more comprehensive fairness analysis:** To address concerns about demographic and genetic biases (Reviewer #2), we have substantially expanded our fairness analysis. We now stratify performance by both self-reported race and genetic ancestry for a more biologically-grounded evaluation of fairness (Fig. R3). Furthermore, we conducted a fine-grained intra-site analysis to disentangle site and race confounding, demonstrating that *FLEX* consistently mitigates bias in both settings (Fig. R4).
- **Strengthened quantitative/statistical evaluation:** In response to feedback on the need for more objective evidence (Reviewer #3), we supplemented UMAP visualizations with objective metrics (LISI and Silhouette Score) and integrated comprehensive statistical testing (paired t-tests and Wilcoxon signed-rank tests with Benjamini-Hochberg correction) across relevant comparisons to confirm significance.
- **Clarified scalability and reliability of the workflow:** To address concerns about the reliance on human experts and GPT-4o (Reviewer #3), we clarified in Methods that expert-guided concept generation is a one-time offline procedure performed during training and is not required at inference, and that the LLM (GPT-4o) is used within a constrained, human-in-the-loop validation pipeline with expert pathologist review.
- **Figure and manuscript presentation improvements:** Based on detailed feedback, all major figures have been revised for clarity (*e.g.*, replaced less informative plot types with boxplots and dumbbell plots where appropriate; improved legends and captions).

We believe that these extensive revisions have significantly improved the clarity, robustness, and impact of our work. We sincerely hope that our responses have fully addressed the your concerns and that the revised manuscript is now suitable for publication in *Nature Communications*.

Thank you again for your time and consideration.

Sincerely,

Authors

Authors' Response to Reviewer 2

General Comments. This manuscript discusses the impacts of foundation models on computational pathology approaches, highlighting biases that are appearing in many of these models. Of note, there are out-of-domain data issues, where models overfit to a subset of the data on which they are trained, and do not generalize well to new tasks, which suggests that these foundation models are not learning the true underlying biology in the pathology slides. The authors perform a comprehensive analysis of some of these biases and proposed a method, FLEX, to correct for them. FLEX is applied to 16 predictive tasks in 4 different TCGA cancer data sets. FLEX shows statistically significant improvement in reducing non-clinically-relevant biases in pathology slide data, notably clinical site biases that are not relevant to the scientific questions the model is being trained to answer but which often confound AI and ML models. This improved method is compared to current standard approaches. The literature is well cited and the manuscript clearly written and easy to follow. Similarly, the figures are beautiful. The work supports the conclusions however additional evidence, outlined in my comments below, would make the claims stronger. As the manuscript is written now, it is unclear if these results will translate to other datasets or if they are specific to TCGA data. There are correlations between clinical sites and other demographic factors that may impact the results, creating relationships between the different sites discussed in the manuscript. The authors do not take these into account when stratifying the data to create IND and OOD subsets of the data. Overall, it is a well written manuscript addressing a significant concern in the field.

Response:

Thank you for your constructive feedback and for acknowledging the clarity of our manuscript and the importance of our work. We have performed new experiments and analyses to address your primary concerns regarding generalizability and the disentanglement of confounding factors.

1. Generalizability Beyond TCGA: To address your concern about the TCGA-specificity of our findings, we conducted zero-shot external validation. We trained our model exclusively on TCGA data and then evaluated its performance on two large-scale, entirely independent external cohorts: the public Clinical Proteomic Tumor Analysis Consortium (CPTAC; $n=1,738$ slides) and our in-house dataset from Nanfang Hospital of Southern Medical University (NFH; $n=5,318$ slides). As detailed in our response to **Comment 6**, FLEX demonstrates significant and consistent performance improvements on both external cohorts (Fig. R6). This provides strong evidence that the model learns fundamental, transferable biological features rather than overfitting to dataset-specific artifacts.

2. Disentangling Site and Demographic Correlations: To address your point about correlations between clinical sites and demographic factors, we have implemented a fine-grained intra-site fairness analysis. As detailed in our response to **Comment 4**, this analysis evaluates model performance and fairness within individual clinical sites, thereby isolating site-specific effects. The results (Fig. R4) demonstrate that FLEX effectively mitigates demographic biases when disentangling site and demographic factors.

We believe these comprehensive revisions have substantially strengthened the manuscript. Detailed point-by-point responses to each of your comments are provided below.

Comment 1

The sites are not defined until the supplemental text. Please make this clearer earlier in the text, that sites reference the literal clinical sites that processing the samples.

Response:

Thank you for this suggestion. We have now explicitly defined this term in the Introduction (Page 2, Line 72) to ensure clarity from the outset.

...These signatures, which stem from variations in tissue preparation, staining protocols, and scanning equipment across different clinical centers (*e.g.*, hospitals or research institutions) where samples are processed (Fig. 1a), directly impede cross-domain generalization...

Comment 2

Based on the results, it looks more like stain normalization works for some but not all sites - what site is out of the norm? Why? Please discuss.

Response:

Thank you for this insightful suggestion. We performed a site-level analysis to investigate the differential effectiveness of stain normalization. Our analysis reveals that the degree of performance improvement correlates with how effectively stain normalization aligns a site's feature distribution with the global average. To investigate this, we quantified the change in image feature deviation for each site relative to the overall features of all sites, both before and after normalization. We then correlated this change with the per-site AUC improvement on two tasks. To measure this "feature deviation", we employed several metrics, including cosine, Euclidean, L2 norm, and Manhattan distances between a site's mean feature vector and the global mean feature vector. A negative change in these metrics signifies that a site's features became more aligned with the global average post-normalization.

For the BRCA-TYPE task (Fig. R1), we observed a statistically significant negative correlation between the change in feature deviation and AUC improvement (*e.g.*, for the 'overall_mean_rel_dev' metric: $r = -0.560, p = 0.037$). This indicates that sites whose features moved closer to the global mean experienced greater performance gains. This finding was replicated in the CRC-TP53 task (Fig. R2), where we again found significant negative correlations (*e.g.*, $r = -0.525, p = 0.030$).

In conclusion, a site is "out of the norm" when stain normalization is less effective at correcting its specific feature-level variations and aligning them with the global data distribution. This analysis quantitatively explains the inconsistent benefits of stain normalization and underscores the need for more robust methods.

Comment 3

On page 6, please reference the supplemental text and tables describing the 16 tasks.

Response:

Thank you for this suggestion. We have added a reference to the supplemental text and tables on page 6 to direct the reader to the detailed descriptions of the 16 predictive tasks.

...Detailed results across all 16 tasks (see 'Dataset and clinical tasks description' in Methods and Supplementary Table 2; Fig. 2d) show consistent performance improvements in both IND and OOD settings...

Comment 4

The racial differences in performance are interesting. TCGA also has information about genetic ancestry (see <https://doi.org/10.1016/j.jccell.2020.04.012>). It would be interesting to see performance differences across genetic ancestry groups, as well as racial stratification. Doing this avoids misclassifying patients with ancestry from multiple human populations (common in African Americans in the TCGA data). This would be a better way to stratify the samples. There are also site-specific race and other demographic factor correlations. How is this handled in the analysis? Are training dataset sizes held constant?

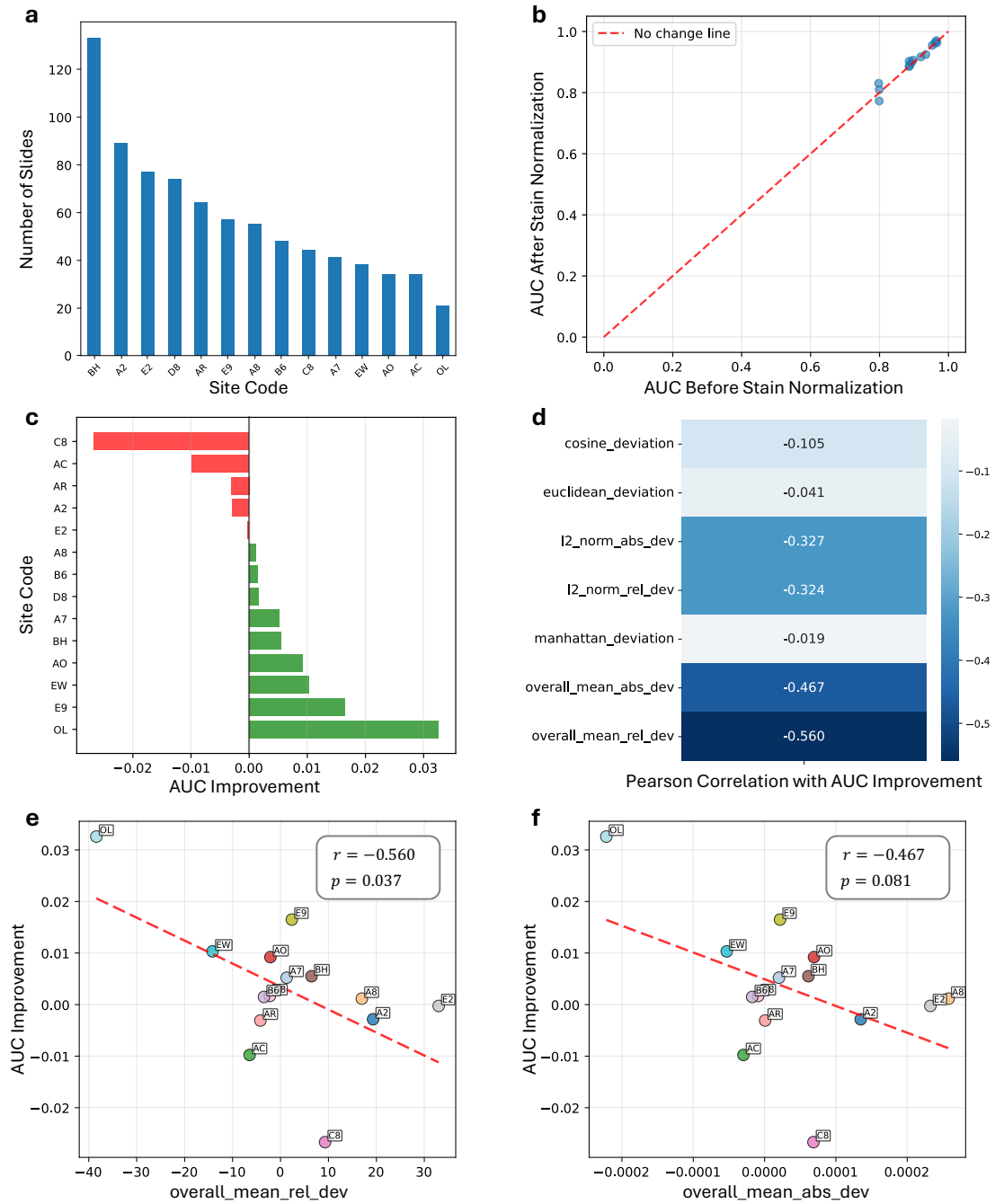


Figure R1: The impact of stain normalization on model performance and its correlation with stain deviation metrics across different acquisition sites in the BRCA-TYPE task. **a**, Distribution of the number of slides for each site included in the analysis (sites with 20 slides). **b**, A scatter plot comparing the site-level AUC before and after stain normalization. Points above the dashed identity line indicate performance improvement. **c**, Bar chart showing the AUC improvement for each site after normalization. Green bars represent an increase in AUC, while red bars represent a decrease. **d**, A heatmap of Pearson correlation coefficients between various stain deviation metrics and the AUC improvement. Darker blue indicates a stronger negative correlation. **e**, **f**, Scatter plots visualizing the relationship between AUC improvement and the two metrics with the strongest negative correlations: (e) `overall_mean_rel_dev` ($r = -0.560, p = 0.037$) and (f) `overall_mean_abs_dev` ($r = -0.467, p = 0.081$).

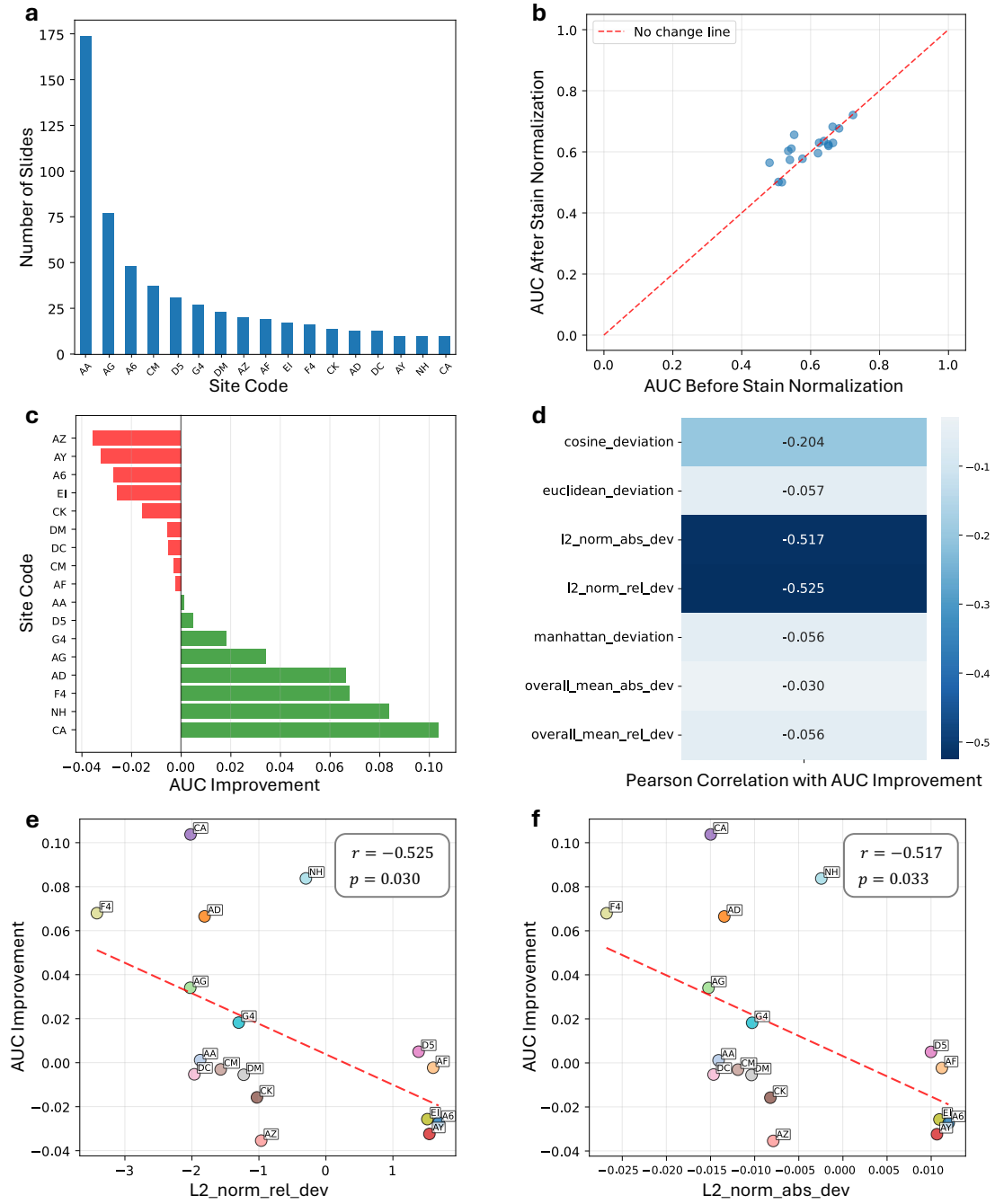


Figure R2: The impact of stain normalization on model performance and its correlation with stain deviation metrics across different acquisition sites in the CRC-TP53 task. **a**, Distribution of the number of slides for each site included in the analysis (sites with 20 slides). **b**, A scatter plot comparing the site-level AUC before and after stain normalization. Points above the dashed identity line indicate performance improvement. **c**, Bar chart showing the AUC improvement for each site after normalization. Green bars represent an increase in AUC, while red bars represent a decrease. **d**, A heatmap of Pearson correlation coefficients between various stain deviation metrics and the AUC improvement. Darker blue indicates a stronger negative correlation. **e**, **f**, Scatter plots visualizing the relationship between AUC improvement and the two metrics with the strongest negative correlations: (e) L2_norm_rel_dev ($r = -0.525, p = 0.030$) and (f) L2_norm_abs_dev ($r = -0.517, p = 0.033$).

Response:

Thank you for this insightful comment and for the valuable suggestions. We agree that using genetic ancestry for fairness evaluation is a more robust approach and that disentangling site-demographic correlations is crucial. In response, we have conducted two new analyses, with results presented in Fig. R3 and R4.

1. Fairness Analysis Using Genetic Ancestry

The racial differences in performance are interesting. TCGA also has information about genetic ancestry (see <https://doi.org/10.1016/j.ccell.2020.04.012>). It would be interesting to see performance differences across genetic ancestry groups, as well as racial stratification. Doing this avoids misclassifying patients with ancestry from multiple human populations (common in African Americans in the TCGA data). This would be a better way to stratify the samples.

Acting on your suggestion, we have expanded our fairness analysis to include stratification by genetic ancestry, providing a more biologically rigorous evaluation alongside self-reported race. As presented in Fig. R3, our analysis confirms that performance disparities exist across genetic ancestry groups with the baseline model. Crucially, FLEX enhances fairness across all ancestry groups. This is demonstrated by the reduced AUC gap ratios (Fig. R3a), more uniform AUC scores across groups (Fig. R3b), and TPR disparity values clustered closer to zero (Fig. R3c-d). This new analysis provides more biologically grounded evidence for FLEX’s effectiveness in promoting demographic fairness. We have also updated the manuscript to reflect this expanded analysis.

— Revision in Page 8, Line 395 —

...Because self-reported race may not fully capture complex biological heritage, we also stratified cohorts by genetic ancestry, which provides a more biologically grounded measure. Our fairness analysis was conducted across both stratification methods in the TCGA dataset to ensure a comprehensive evaluation....

2. Disentangling Site-Specific and Demographic Correlations

There are also site-specific race and other demographic factor correlations. How is this handled in the analysis? Are training dataset sizes held constant?

We agree that the correlation between clinical sites and demographic factors is a significant confounding variable. To address this, we conducted a multi-faceted analysis to demonstrate that FLEX mitigates demographic bias independently of site-level effects.

First, we performed a fine-grained intra-site fairness analysis to disentangle these variables (Fig. R4). This analysis provides evidence from the following two perspectives:

- **Consistent Performance Across Sites:** We first confirmed that FLEX improves performance across all major sites, irrespective of their demographic compositions (Fig. R4b).
- **Intra-Site Fairness:** Crucially, when evaluated within individual sites, FLEX still significantly reduces the performance gap between White and Black patient groups (Fig. R4c-f).

This intra-site analysis demonstrates that FLEX’s fairness improvements are not merely a byproduct of correcting site-level batch effects but stem from a more fundamental mitigation of demographic bias.

Second, we confirmed these findings generalize beyond TCGA’s unique confounding structure. Our zero-shot validation on two independent external cohorts (CPTAC and NFH, Fig. R3 and Fig. R6) showed that a model trained only on TCGA maintains superior performance and fairness on these OOD datasets. This further validates that FLEX learns generalizable biological features rather than relying on dataset-specific artifacts or confounds.

Finally, this empirical success is rooted in FLEX’s core design. FLEX uses an information bottleneck guided by textual pathology concepts (*e.g.*, “invasive ductal carcinoma”) that are inherently free of site or demographic information. This architecture constrains the model to learn features aligned with these unbiased concepts and to discard spurious correlations, including entangled site and demographic signatures. We have updated the manuscript to clarify this mechanism.

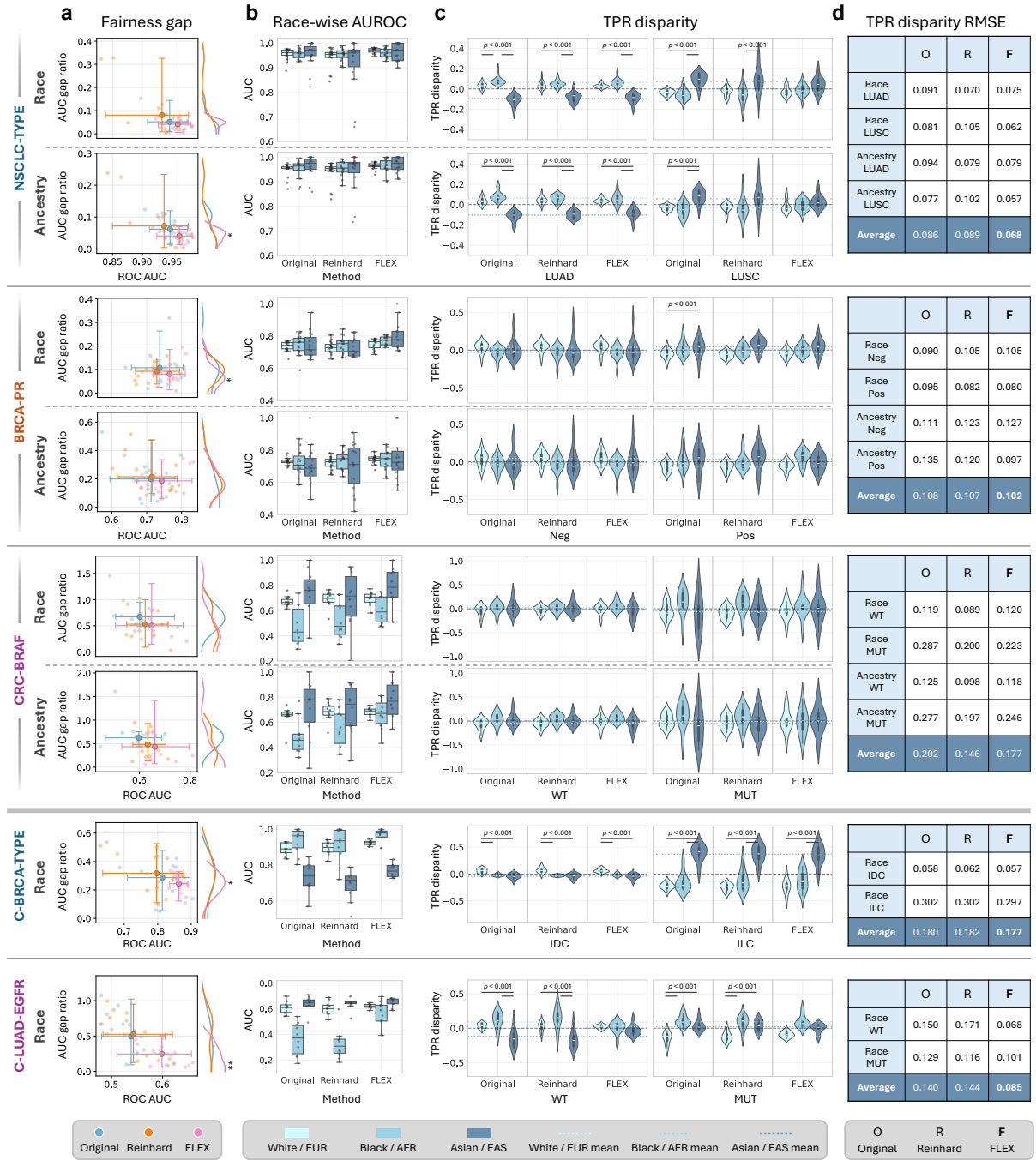


Figure R3: (Figure 3 in manuscript) Evaluation of FLEX's effectiveness in improving demographic fairness. Fairness is evaluated across race and ancestry groups using results from 15 cross-validation folds. For each fold, metrics for each demographic-label group were estimated via bootstrapping (500 resamples). **a**, The fairness gap for five representative tasks. The AUC gap ratio is the performance difference between the best- and worst-performing subgroups relative to the overall AUC. Smaller dots represent individual folds; large dots with error bars show the mean and 95% CI. A lower ratio indicates better fairness. Marginal distributions show the distribution of AUC gap ratios across folds. Statistical significance ($*p < 0.05$, $**p < 0.01$) from a paired Wilcoxon signed-rank test denotes the improvement of FLEX over the Original method. **b**, The AUROC distribution across subgroups. Boxes indicate the interquartile range of the AUROC distribution ($n = 15$ folds), with the center being the 50th percentile and the whiskers extending to data points within $1.5 \times$ the interquartile range. Tighter distributions indicate more equitable performance. **c**, Violin plots of True Positive Rate (TPR) disparity, defined as the difference between a subgroup's TPR and the overall TPR. Distributions centered closer to the zero-disparity line indicate higher fairness. P-values from the paired t-test are shown. The dashed line represents the overall TPR across three methods. **d**, Quantitative summary of TPR disparity, showing the Root Mean Square Error (RMSE) for Original (O), Reinhard (R), and FLEX (F). Lower values indicate better fairness.



Figure R4: (Supplementary Figure 5 in manuscript) Disentangling site and demographic biases: FLEX mitigates racial performance gaps within individual clinical sites. **a**, Distribution of slides by self-reported race across all TCGA sites, highlighting demographic imbalances and site-specific correlations. **b**, Overall performance (AUROC) comparison on the BRCA-TYPE task across eight major clinical sites (slides > 5 per class). FLEX consistently outperforms baseline methods. Statistical significance is denoted by asterisks ($*p < 0.05$, $**p < 0.01$, $***p < 0.001$) based on a paired t-test between FLEX and the second-best method. **c-f**, Fine-grained fairness analysis within four of the major individual clinical sites (BH, E2, B6, EW). For each site, the analysis includes: **Sample distribution table**: Detailed patient counts stratified by race (White, Black) and tumor subtype (IDC, ILC). **Fairness gap plot**: Visualizes the AUC gap (absolute performance difference between White and Black patients) against overall site-level AUROC. Small dots represent individual cross-validation folds, while large dots with error bars show the mean and 95% CI. A lower ratio (closer to the y-axis) indicates better fairness. Statistical significance ($p < 0.05$) from a paired Wilcoxon signed-rank test denotes the improvement of FLEX over the Original method. **Race-wise AUROC plot**: Box plots showing the AUROC distribution for White and Black patient groups. Tighter and more aligned distributions indicate more equitable performance. Boxes show Q1-Q3 interquartile ranges, with the center line indicating the median and whiskers extending to $1.5 \times$ the interquartile range. **TPR disparity plot**: Violin plots of True Positive Rate (TPR) disparity, defined as the difference between a subgroup's TPR and the site's overall TPR. Distributions centered closer to the zero-disparity line indicate higher fairness. **TPR disparity RMSE table**: A quantitative summary of TPR disparity, showing the Root Mean Square Error (RMSE) for Original (O), Reinhard (R), and FLEX (F). Lower values indicate better fairness.

...FLEX represents a significant step in adapting foundation models for computational pathology. It effectively mitigates the negative impacts of site-specific signatures and demographic biases while improving feature discriminability for downstream tasks. The core mechanism of FLEX uses an information bottleneck guided by textual pathological concepts to disentangle task-relevant features from spurious variations. Because these textual concepts are inherently free of site or demographic information, this process forces the model to discard extraneous signals, including complex site and demographic signatures. The efficacy of FLEX was validated through extensive evaluations across 16 pathology tasks, using various pathology VLMs and MIL frameworks. These consistent results, obtained via a site-preserved cross-validation strategy and supported by fairness analyses, underscore the method’s robustness. Beyond performance improvements, FLEX offers insights into the strategic use of multimodal prior knowledge for task-specific adaptation of foundation models, contributing to the development of more effective and equitable diagnostic tools....

Regarding training data sizes, they were not held constant. We used all available training data in each fold of our site-preserved cross-validation. However, the partitioning strategy uses convex optimization to ensure that each site-fold contains a balanced proportion of patients for the diagnostic labels [1], resulting in roughly comparable training set sizes across folds (Table R1). Furthermore, our ablation study (Fig. 5 in manuscript) demonstrates that FLEX consistently improves performance across different training data scales.

Comment 5

Several of the biomarker prediction tasks (*e.g.*, BRCA HER2, ER, and PR) have known racial biases. Others are not clearly known. Please discuss the variations in performance given known biases and performance improvements relative to the tasks - is there a larger performance gain in tasks where there are known racial biases?

Response:

Thank you for this valuable suggestion. We performed a quantitative comparison across biomarker prediction tasks with well-documented racial disparities (*e.g.*, BRCA-HER2, BRCA-ER, BRCA-PR) and tasks where such disparities are less established (*e.g.*, STAD-EBV, STAD-MSI). As shown in Fig. R5, our analysis did not reveal a clear correlation between the magnitude of improvement from FLEX and the presence of known clinical racial disparities.

This finding, while potentially counter-intuitive, suggests several potential insights into the nature of bias in computational pathology and the mechanism of FLEX.

First, in datasets like TCGA, biases from data acquisition (“site-specific signatures”) are heavily confounded with demographic factors. Our results suggest that these procedural, data-driven artifacts may be a more dominant source of model error than the currently understood biological differences. FLEX excels at correcting these pervasive data-driven biases.

Second, FLEX operates by aligning visual features with a bias-free textual concept space. This mechanism acts as a filter for any spurious visual correlation not explicitly related to the core biological concept. Its debiasing effect is therefore inherently general and agnostic to the source of bias, rather than being specifically tuned to one type.

Third, clinically reported racial disparities often reflect macroscopic phenomena. It is plausible that these macro-level differences do not always manifest as distinct, learnable microscopic morphological features on a pathology slide, which could weaken the expected correlation.

While our current analysis does not establish a clear link between known clinical biases and the extent of performance gain, the interplay between biological and data-driven biases is undoubtedly complex. We have marked this as an important area for future investigation.

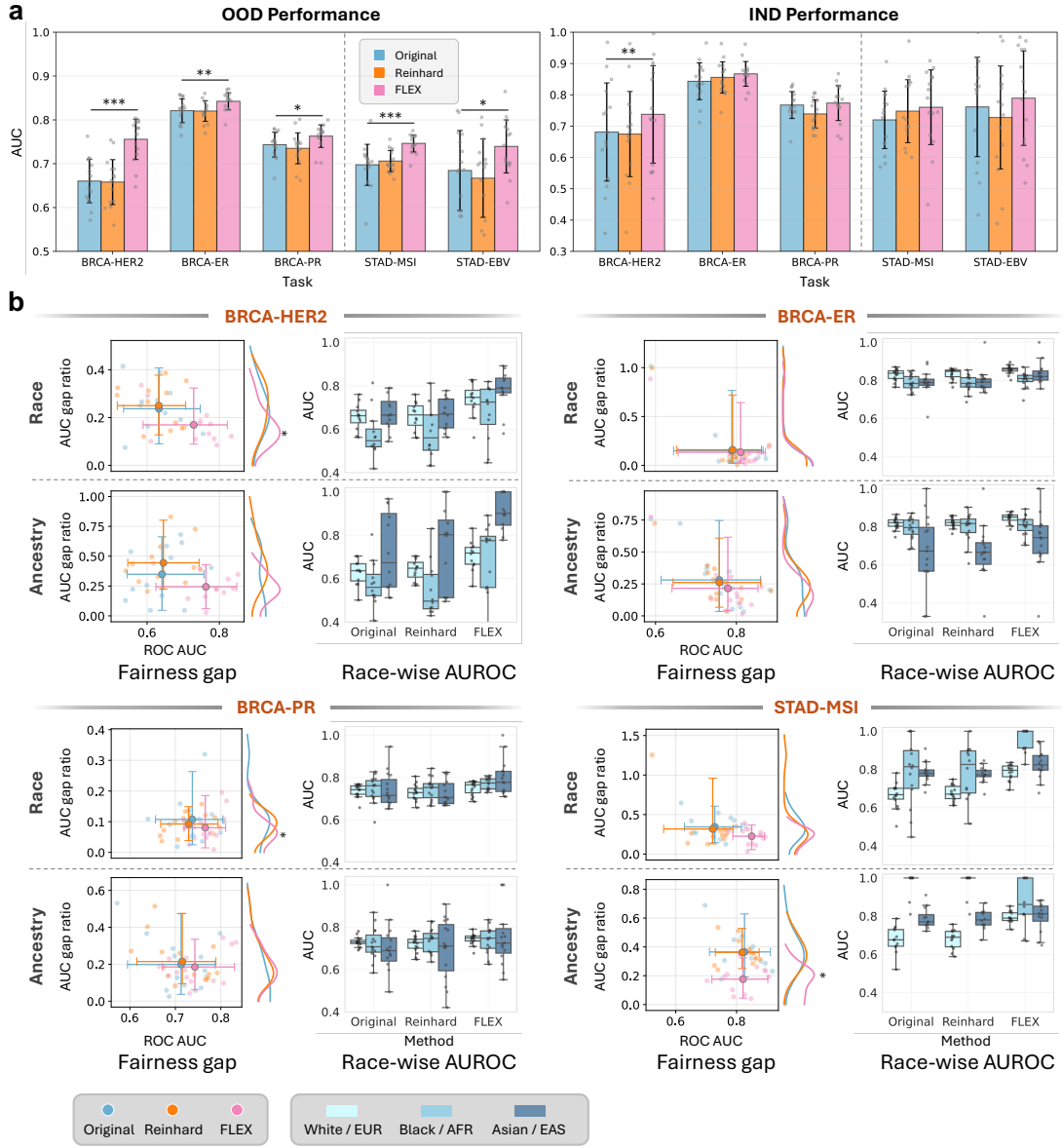


Figure R5: (Supplementary Figure 6 in manuscript) Analysis of FLEX’s performance and fairness improvements on biomarker prediction tasks, stratified by self-reported race and genetic ancestry. a, Bar plots comparing Out-of-Domain (OOD) and In-Domain (IND) AUROC performance of the original features, features after Reinhard stain normalization, and features enhanced by FLEX across five biomarker prediction tasks. FLEX consistently improves diagnostic performance, with asterisks indicating statistical significance of the improvement compared to the original model (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). Error bars represent 95% CI. **b,** Detailed fairness analysis for four representative tasks, stratified by self-reported race (top row for each task) and genetic ancestry (bottom row for each task). **Fairness gap plots** (left): The y-axis represents the fairness gap (performance difference between the best- and worst-performing groups), and the x-axis shows the overall AUROC. The goal is to move towards the bottom-right corner (lower gap, higher performance). **Race/Ancestry-wise AUROC plots** (right): Box plots showing the AUROC distribution for White/European (EUR), Black/African (AFR), and Asian/East Asian (EAS) groups. The results demonstrate that FLEX consistently reduces the fairness gap and mitigates performance disparities across demographic groups, for both self-reported race and the more granular genetic ancestry stratification. Boxes show Q1-Q3 interquartile ranges, with the center line indicating the median and the whiskers extending to data points within $1.5\times$ the interquartile range.

Comment 6

It's unclear how much of the model performance is dependent on dataset biases beyond demographic data. Ideally, the trained model would be applied to non-TCGA and racial bias and other biases analyzed in this way. Other datasets exist with pathology slide images, and models trained in this manuscript could be applied to those.

Response:

Thank you for this crucial point. To address this, we conducted a zero-shot external validation. We trained FLEX exclusively on TCGA data and then evaluated its performance on two large, independent external cohorts: the public CPTAC dataset ($n=1,738$, three cancer types) and a large in-house dataset from Nanfang Hospital of Southern Medical University (NFH, $n=5,318$, four cancer types). These datasets introduce new sources of variation, including different patient populations, slide preparation protocols, and scanners, providing a stringent test of out-of-distribution generalization.

The results (Fig. R6) demonstrate that FLEX significantly enhances the model's ability to generalize to these unseen datasets. As summarized in Fig. R6a, the TCGA-trained model with FLEX achieves a higher average AUC across all tasks on both CPTAC (0.714 vs. 0.679 for baseline, $p < 0.001$) and NFH (0.771 vs. 0.733, $p < 0.001$) compared to the original model. This indicates a consistent improvement in out-of-distribution generalization. The radar plot (Fig. R6b) and detailed box plots (Fig. R6c, R6d) demonstrate that FLEX consistently expands the performance frontier across most tasks in both external cohorts.

These findings provide strong evidence that the performance gains from FLEX are not an artifact of overcoming biases specific to TCGA. Instead, FLEX enables the model to learn more fundamental and robust biological features, allowing it to generalize effectively across the significant domain shifts inherent in new clinical datasets. This demonstrates that FLEX is a valuable tool for building more reliable and generalizable models for real-world application.

Comment 7

Comments on Figure 2:

1. Why is figure 2 referenced before figure 1?
2. Figure 2 plots should have axis labels.
3. In Figure 2, UMAP is a visualization technique that is affected by parameter tuning. How much of that is causing the biases here?
4. How are the site stratifications actually done, as visualized in Fig 2b? This visualization doesn't explain it well. How were sites grouped? What about site-site relationships? For example, is there a relationship between the clinical equipment, the location, or something else?

Response:

Thank you for the detailed feedback on Figure 2. We have revised the figure and manuscript to address these points.

1. Why is figure 2 referenced before figure 1?

We have corrected the text to ensure figures are referenced in order.

2. Figure 2 plots should have axis labels.

We have added axis labels to all plots in Figure 2.

3. In Figure 2, UMAP is a visualization technique that is affected by parameter tuning. How much of that is causing the biases here?

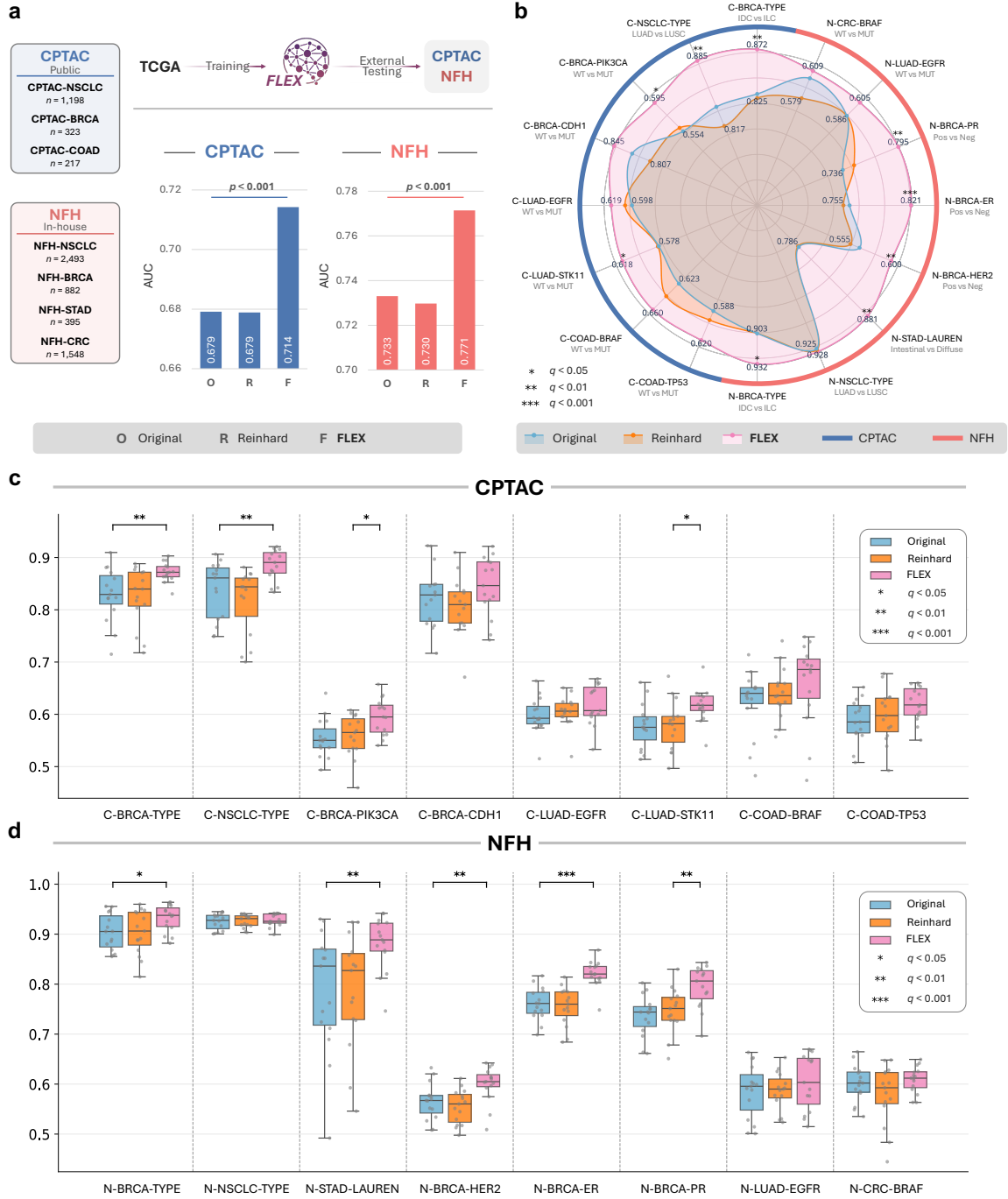


Figure R6: Zero-shot generalization of TCGA-trained models to external CPTAC and in-house NFH cohorts. Models were trained exclusively on TCGA data and evaluated directly on two independent external datasets without fine-tuning. **a**, Overall experimental design and results. Bar plots display the average AUROC across all tested tasks on the CPTAC (left) and in-house NFH (right) datasets. FLEX significantly improves the average performance of the TCGA-trained model on both external cohorts ($p < 0.001$, paired t-test). “O”, “R”, and “F” denote Original, Reinhard, and FLEX, respectively. **b**, Radar plot comparing task-specific AUROC across models on CPTAC and NFH data. FLEX consistently expands the performance radius across tasks. Asterisks indicate statistical significance of the improvement by FLEX compared to the original model (* $q < 0.05$, ** $q < 0.01$, *** $q < 0.001$, Benjamini-Hochberg FDR correction). **c**, **d**, Detailed box plots showing task-by-task performance on the (c) CPTAC and (d) NFH cohorts. FLEX consistently outperforms or matches the baseline and Reinhard-normalized models, demonstrating its robust ability to enhance zero-shot transfer learning by mitigating dataset-specific biases. Boxes indicate quartile values of the AUROC distribution ($n = 15$ folds), with the center line indicating the median and whiskers extending to $1.5\times$ the interquartile range.

Thank you for raising this important point. In the manuscript, we set UMAP parameters to ‘n_neighbors=30’ and ‘min_dist=0.1’ based on common practice and preliminary experiments to balance local and global structure. To ensure that our conclusions are not artifacts of specific UMAP settings, we conducted a sensitivity analysis by varying these hyperparameters. As shown in Fig. R7, the results for both Original and Reinhard features show strong site-specific clusters across all tested parameter settings, demonstrating that the observed bias is an inherent property of these feature spaces. In contrast, features enhanced by FLEX consistently show a well-mixed distribution regardless of the UMAP parameters, confirming the robustness of our method’s effect.

4. How are the site stratifications actually done, as visualized in Fig 2b? This visualization doesn’t explain it well. How were sites grouped? What about site-site relationships? For example, is there a relationship between the clinical equipment, the location, or something else?

Thank you for this essential question. We appreciate the opportunity to provide a more detailed clarification. Our site splitting is based on the established Preserved-Site Cross-Validation technique [1], [2], which is specifically designed for a stringent evaluation of out-of-distribution (OOD) generalization. The principle is to test if a model can perform well on data from a source it has never seen.

Therefore, we do not group sites based on partial information like equipment or location. The true “site-specific signature” is a complex and often undocumented amalgamation of numerous factors (scanner calibration, staining reagents, pre-analytic procedures). Attempting to group sites based on a few known variables would fail to capture this full spectrum of potential bias.

Instead, our method treats each site as an independent entity. We partition the entire set of TCGA sites into distinct folds for cross-validation. To ensure a stable evaluation, we use the convex optimization strategy from Howard et al. [1] to balance the patient-level diagnostic labels across each fold.

This process guarantees that each OOD test set is composed of sites entirely unseen during training while having a label distribution comparable to the training data, providing a robust assessment of a model’s ability to generalize to new data sources.

Comment 8

Comments on Figure 3:

1. Please move figure legends under the matching figures.
2. It’s unclear exactly what Figure 3a is plotting. Please provide more clarity in the caption description.
3. Please add a horizontal line for the avg performance in the data overall, in Figure 3b.
4. Plots used in Figure 3c are not appropriate for the data being modeled. Please update these to not use dynamite plots.
5. For all of figure 3, please define the error bars and other data annotations for all subfigures.

Response:

We appreciate the specific suggestions for improving Figure 3 and have made the following revisions:

1. Please move figure legends under the matching figures.

The legends have been moved to be placed directly under their corresponding subfigures.

2. It’s unclear exactly what Figure 3a is plotting. Please provide more clarity in the caption description.

We have rewritten the caption for Figure 3a to provide a more precise description.

3. Please add a horizontal line for the avg performance in the data overall, in Figure 3b.

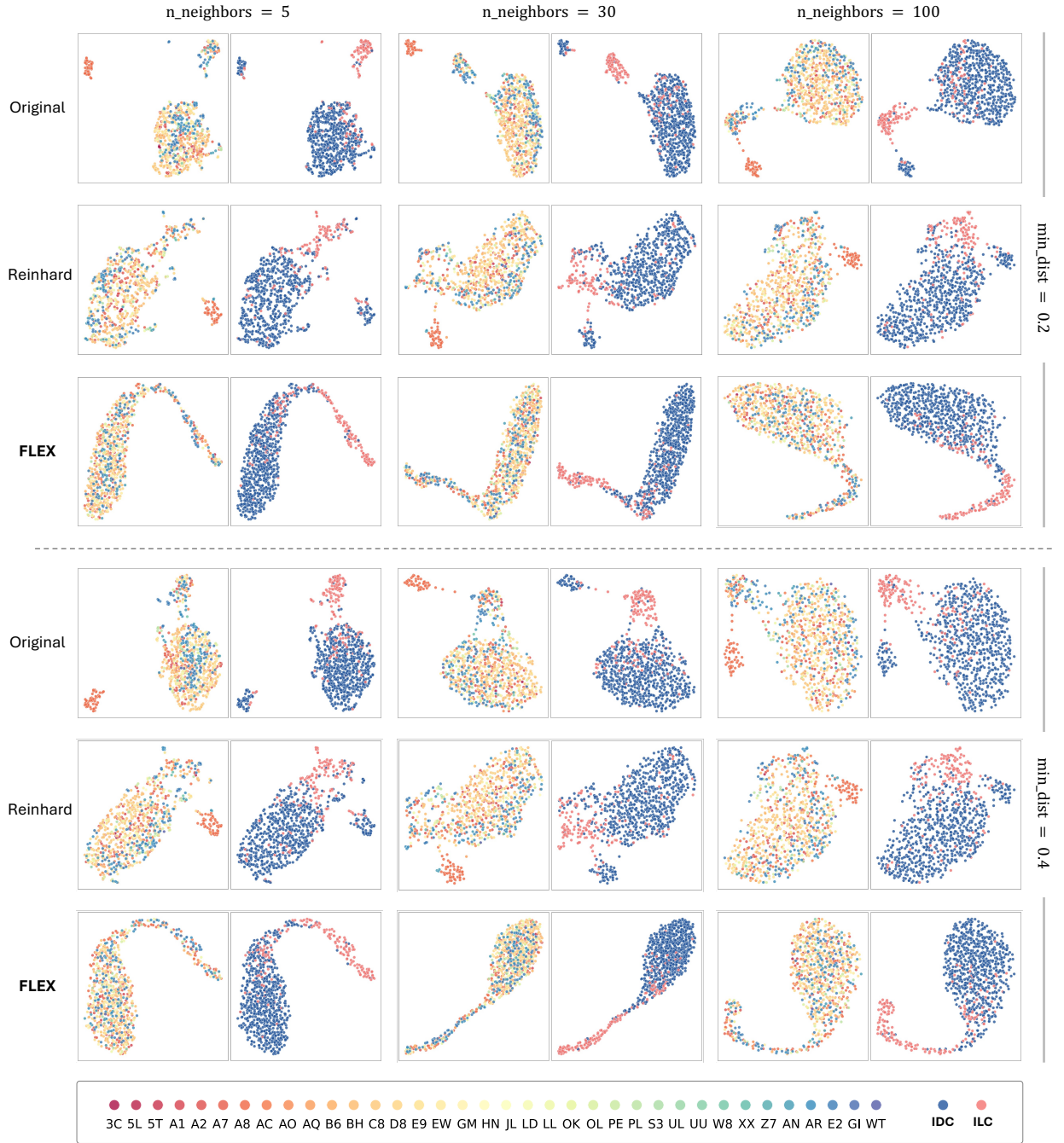


Figure R7: (Supplementary Figure 4 in manuscript) Robustness analysis of UMAP visualizations to hyperparameter tuning. Parameters ‘ $n_neighbors$ ’ and ‘ min_dist ’ were varied to assess their impact on the UMAP representation of different feature types. The original and Reinhard-normalized features show strong site-specific clustering (color-coded by site) across all parameter settings. In contrast, features enhanced by FLEX consistently exhibit a well-mixed distribution, indicating effective mitigation of site-specific signatures.

We have added a horizontal line representing the average performance to provide a clear baseline, as shown in Fig. R3c. We have also added a quantitative summary of TPR disparity in Fig. R3d, showing the Root Mean Square Error (RMSE) of TPR disparity values across demographic groups for each method. This provides an objective measure of fairness, demonstrating that FLEX achieves the lowest average RMSE across most tasks.

4. Plots used in Figure 3c are not appropriate for the data being modeled. Please update these to not use dynamite plots.

We have replaced them with box plots to better represent the data distribution.

5. For all of figure 3, please define the error bars and other data annotations for all subfigures.

We have now clearly defined all error bars and other annotations in the figure captions.

Comment 9

Comments on Figure 4:

1. Please do not use dynamite plots for Figure 4a. Box plots, violin plots, or another visualization are a more representative way of visualizing these results.
2. Please update the y-axis ranges to be consistent across each row in Figure 4a so it is easier to compare model performance across tasks.
3. It's difficult to see the EBV +/- shapes in the Figure 4b plots. Please adjust to make these easier to distinguish.

Response:

Thank you for the constructive feedback on Figure 4. We have updated the figure as follows:

1. Please do not use dynamite plots for Figure 4a. Box plots, violin plots, or another visualization are a more representative way of visualizing these results.

We have replaced the plots with box plots for a more representative visualization.

2. Please update the y-axis ranges to be consistent across each row in Figure 4a so it is easier to compare model performance across tasks.

We have adjusted the y-axis ranges to be consistent across each row in Figure 4a.

3. It's difficult to see the EBV +/- shapes in the Figure 4b plots. Please adjust to make these easier to distinguish.

Thank you for this constructive feedback. We have revised the figure to improve readability and provide stronger, quantitative evidence for our claims, as shown in Fig. R8. We now present two parallel UMAP subplots for each method: one colored by clinical site (to assess batch effect removal) and one colored by EBV status (to evaluate class separability). This faceted presentation is much clearer.

More importantly, to provide objective proof, we have included two key metrics calculated on the high-dimensional features: the Local Inverse Simpson's Index (LISI) and the Silhouette Score. A higher LISI score indicates better mixing of sites (weaker batch effects), while a higher Silhouette Score indicates better-defined class clusters. These metrics provide quantitative support for the visual improvements.

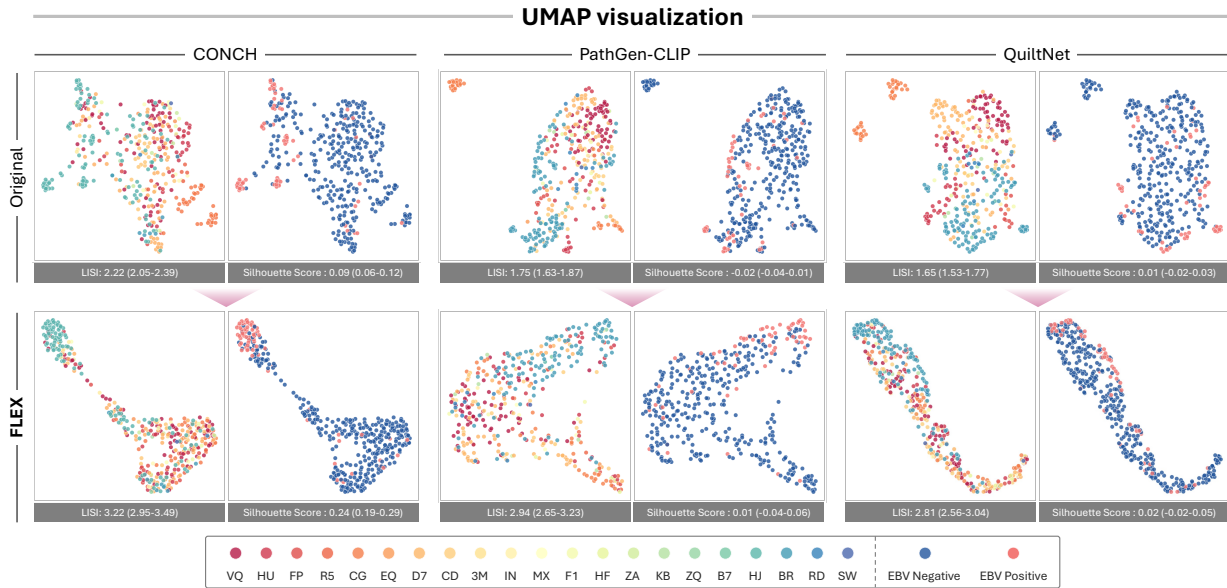


Figure R8: (Figure 4b in manuscript) UMAP visualizations illustrating the effect of FLEX on the patch feature space for the STAD EBV task. For each VLM, parallel subplots are colored by site (left) to visualize batch effect mitigation, and by EBV status (right) to visualize class separability. The Local Inverse Simpson's Index (LISI) for site integration (higher is better) and the Silhouette Score for class separation (higher is better) provide quantitative evidence. The visualizations and scores demonstrate that FLEX reduces site-specific clustering while improving the discriminability of task-relevant classes.

Comment 10

Comments on Figure 5:

- Figure 5a missing a legend and the statistical tests is not defined. Are these adjusted for multiple hypotheses? Also, are the data repeatedly stratified then models tested? How much overlap is there in data between each test, and how is the significance of performance differences calculated? What size is the training data for each of these splits?
- Figure 5b would be more impactful if visualized differently, to show the change in performance across original, improvement, and FLEX. Perhaps using a bipartite graph.
- Additionally, there seems to be a large amount of variability in performance in Figure 5b, suggesting that there are other factors in the data splits driving performance differences. Please discuss this in the text.

Response:

Thank you for this detailed feedback on Figure 5. We have made substantial revisions to the figure and its caption to clarify the experimental design and results.

- We have revised the figure and caption to address your points, as shown in the new Fig. R9.

1.1: Figure 5a missing a legend and the statistical tests is not defined. Are these adjusted for multiple hypotheses?

We have added a legend and clarified the statistical tests in the caption. We performed paired t-tests for performance comparisons and adjusted p-values for multiple hypotheses using the Benjamini-Hochberg FDR procedure. Asterisks in the figure indicate significance levels.

1.2: Also, are the data repeatedly stratified then models tested?

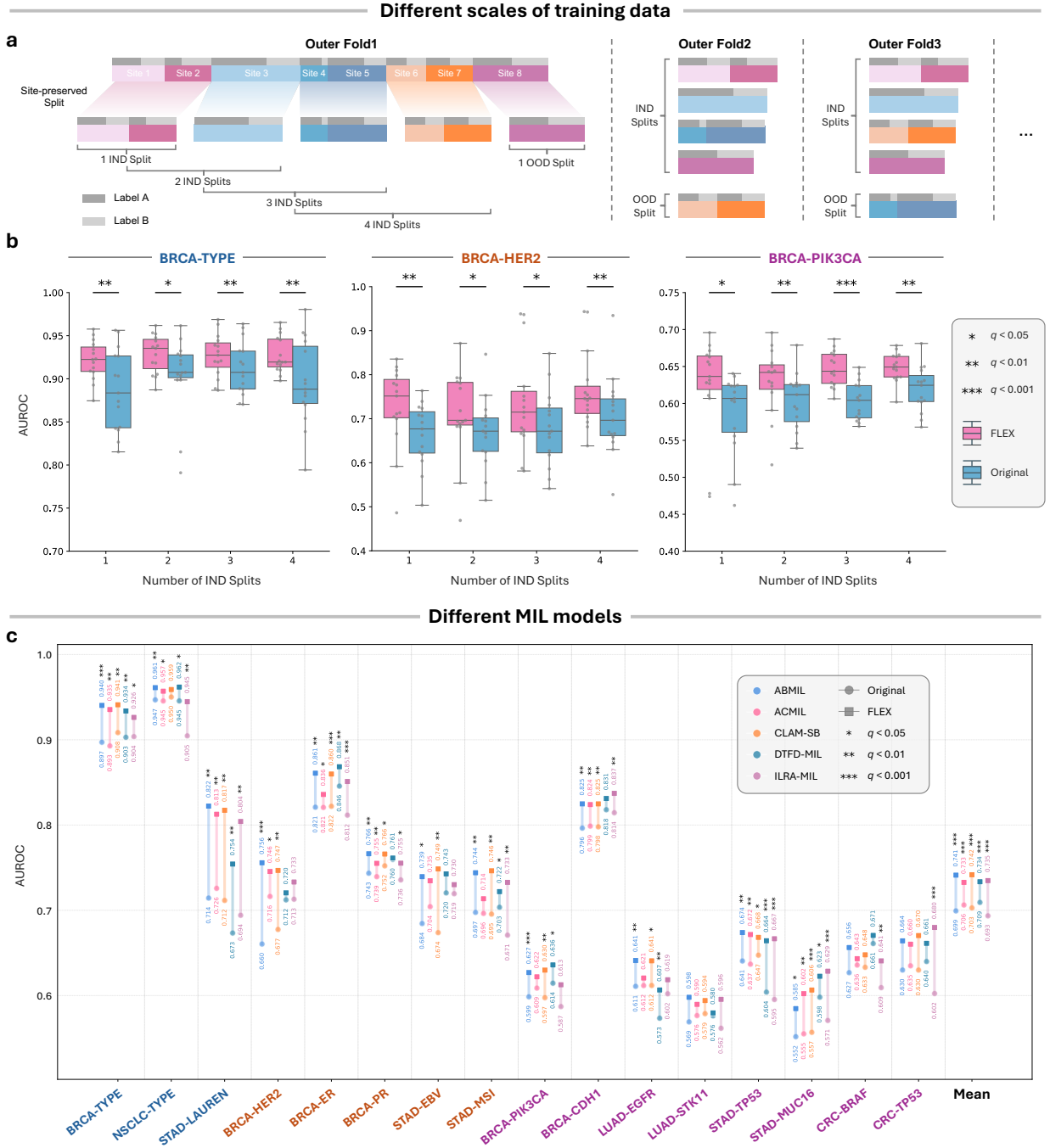


Figure R9: (Figure 5 in manuscript) Performance of FLEX across training data scales and MIL models. **a**, Schematic of the experimental design for evaluating performance at different training data scales. The procedure follows our main Site-Preserved Monte Carlo Cross-Validation (SP-MCCV) strategy, which involves 5 outer folds and 3 inner Monte Carlo Cross-Validation (MCCV) folds, for a total of 15 evaluation runs. Specifically for this experiment, the dataset is first partitioned into 5 site-preserved outer folds. In each of the 5 outer loop iterations, one fold is held out as the OOD test set. Training sets of increasing size (1, 2, 3, and 4 folds) are constructed from the remaining 4 folds to create a learning curve. For each scale condition, 3 inner Monte Carlo Cross-Validation (MCCV) runs are performed to ensure stable results, for a total of 15 evaluation runs per condition. **b**, OOD AUROC comparison between FLEX and baseline (Original). Box plots show aggregated results from 15 runs for each training data scale. Asterisks indicate statistically significant improvement (Benjamini-Hochberg adjusted q -value: $* < 0.05$, $** < 0.01$, $*** < 0.001$). Box plots show Q1–Q3 interval with the median; whiskers are $1.5 \times$ interquartile range. **c**, Dumbbell plot showing OOD AUROC after integrating FLEX with five state-of-the-art MIL models across 16 tasks. FLEX consistently improves performance across different MIL architectures.

Yes, the results are based on a repeated stratification process. Our experimental design (Fig. R9a) follows a Site-Preserved Monte Carlo Cross-Validation (SP-MCCV) strategy with 5 outer folds and 3 inner Monte Carlo Cross-Validation (MCCV) folds, for a total of 15 evaluation runs. Specifically, we partition the data into 5 mutually exclusive, site-preserved folds. In each of the 5 iterations of the outer loop, one fold is held out as the out-of-domain (OOD) test set. For each training data scale (1, 2, 3, or 4 IND splits), we perform 3 repeated runs (inner MCCV folds) to ensure the stability and robustness of the results. This process results in a total of 15 evaluation runs (5 outer folds $\times$ 3 inner runs) for each data scale condition. The box plots in Fig. R9b show the aggregated distribution of these 15 results.

1.3: How much overlap is there in data between each test...?

The overlap between each test can be summarized as follows:

- **Between each outer fold:** There is no overlap between the OOD test sets used in each of the 5 outer loop iterations. They are, by design, mutually exclusive partitions. This ensures that our overall performance metric is an average over multiple, independent unseen test sets.
- **Within an outer fold (across different data scales):** Within a single outer loop iteration, the OOD test set is fixed, and the training sets are intentionally nested and overlapping. For example, the training set for “2 IND splits” is a superset of the training set for “1 IND split”. Because of this dependency, our primary statistical claim focuses on the comparison between FLEX and the baseline at each data scale, rather than comparing performance across different scales.

1.4: ...and how is the significance of performance differences calculated?

The significance was calculated using a paired t-test on the vector of 15 AUROC scores from our 5 $\times$ 3 nested cross-validation for each method. The resulting p-values were then adjusted using the Benjamini-Hochberg FDR procedure.

1.5: What size is the training data for each of these splits?

We have included a new table (Table R1) in the Supplementary Materials detailing the training and OOD test set sizes for each task and data scale condition, which provides a clear overview of how many samples were used for training and testing at each stage of our evaluation.

Table R1: Training and OOD Test Set Sizes for Each Task and Data Scale Condition. The table shows the number of samples in the training set for each task at different data scales (1 to 4 IND splits) and the corresponding OOD test set sizes. O1 to O5 represent the five outer folds in the Site-Preserved Monte Carlo Cross-Validation (SP-MCCV) strategy.

Task	Number of IND Splits	Training Set Size (Samples)					OOD Test Set Size (Samples)				
		O1	O2	O3	O4	O5	O1	O2	O3	O4	O5
BRCA-TYPE	1	117	117	118	135	123	186	174	178	190	209
	2	235	237	254	261	242	186	174	178	190	209
	3	353	370	376	376	357	186	174	178	190	209
	4	491	501	493	496	480	186	174	178	190	209
BRCA-HER2	1	125	140	123	140	125	178	179	196	176	198
	2	262	259	259	261	250	178	179	196	176	198
	3	384	399	389	390	386	178	179	196	176	198
	4	527	527	515	530	512	178	179	196	176	198
BRCA-PIK3CA	1	123	123	136	124	135	197	179	177	196	178
	2	249	262	264	258	258	197	179	177	196	178
	3	387	387	399	391	389	197	179	177	196	178
	4	511	530	528	510	526	197	179	177	196	178

2. Figure 5b would be more impactful if visualized differently, to show the change in performance across original, improvement, and FLEX. Perhaps using a bipartite graph.

Thank you for this suggestion. Following your advice, we have redesigned the figure as a Dumbbell Plot (Fig. R9c). For each task and MIL model, a connecting line explicitly shows the performance change from baseline to FLEX. The length and direction of each line intuitively represent the magnitude of the performance gain. Asterisks are used to indicate the statistical significance of the performance improvement, with all p-values adjusted for multiple hypotheses using the Benjamini-Hochberg FDR procedure. We believe this format is more direct and impactful in demonstrating the consistent improvements our method delivers across different models and tasks.

3. Additionally, there seems to be a large amount of variability in performance in Figure 5b, suggesting that there are other factors in the data splits driving performance differences. Please discuss this in the text.

Thank you for this observation. We have now added a discussion of this phenomenon to the manuscript:

— Revision in Page 10, Line 530 —

...The performance improvement seen in Fig. 5c varies across the 16 pathology tasks due to three main factors: intrinsic task difficulty, the utility of feature refinement for a given task, and the complexity of the MIL model. First, inherent task difficulty sets the upper bound on performance; for example, morphology classification tasks consistently yield higher AUROCs than most gene mutation prediction tasks. Second, the magnitude of improvement from FLEX depends on task-specific conditions. In the STAD-LAUREN task, for instance, all MIL models show a substantial performance gain. This may be because the TCGA-STAD cohort is relatively small, causing baseline models to overfit to noisy features. FLEX’s core mechanism—refining features by filtering task-irrelevant information—improves the signal-to-noise ratio, which is particularly beneficial when training data is limited. Third, performance gains are more pronounced with simpler MIL models like ABMIL and CLAM-SB compared to more complex architectures. We hypothesize this is related to the joint optimization of FLEX and the MIL model. A simpler MIL backbone may create a more stable learning environment, allowing gradients to effectively guide the feature refinement process within FLEX. In contrast, with more intricate MIL models, the increased number of parameters and interacting components can complicate convergence to an optimal state, potentially limiting the observable performance gain from FLEX....

References:

- [1] F. M. Howard, J. Dolezal, S. Kochanny, *et al.*, “The impact of site-specific digital histology signatures on deep learning model accuracy and bias,” *Nature communications*, vol. 12, no. 1, p. 4423, 2021.
- [2] A. Vaidya, R. J. Chen, D. F. Williamson, *et al.*, “Demographic bias in misdiagnosis by computational pathology models,” *Nature Medicine*, vol. 30, no. 4, pp. 1174–1190, 2024.

Authors’ Response to Reviewer 3

General Comments. The proposed method presents an interesting model design. There are a few potential issues regarding its scalability, interpretability, and practical utility. Notable limitations include reliance on human experts for patch selection, dependence on GPT-4o for concept generation without verification, and modest improvements in out-of-domain performance. In addition, the absence of statistical testing across tasks may undermine the method’s robustness and its real-world applicability.

Response:

Thank you for your comprehensive feedback. We have substantially revised the manuscript to address your concerns regarding scalability, reliance on external knowledge, interpretability, performance, and statistical robustness.

1. Scalability and Reliance on External Knowledge: We have clarified in the manuscript that knowledge integration is a one-time, offline procedure conducted solely during model training. The patch selection process is efficient, requiring expert validation on only a small number of patches and thereby minimizing the demand on expert time. This setup cost does not impact the scalability or speed of the final model at inference. To address the concern about unverified AI outputs, we have detailed our human-in-the-loop validation protocol and constrained fact-based concept generation strategy to ensure clinical reliability (as detailed in our responses to **Comments 1 and 2**).

2. Interpretability: To improve clarity, we have provided a more precise definition for “noisy regions” in our attention map visualizations, defining them as areas containing processing artifacts or task-irrelevant biological structures (as detailed in our response to **Comment 6**).

3. OOD Performance and Practical Utility: To address concerns about the magnitude of OOD performance improvements and practical utility, we conducted an extensive zero-shot external validation on two large, independent cohorts: the public Clinical Proteomic Tumor Analysis Consortium collections (CPTAC; $n=1,738$) and our in-house dataset from Nanfang Hospital of Southern Medical University (NFH; $n=5,318$). The results (Fig. R11c, R15) show that FLEX yields statistically significant performance gains in these challenging OOD scenarios, providing strong evidence of the method’s robust and generalizable benefits for real-world applications (as detailed in our responses to **Comments 3 & 10**).

4. Statistical Testing: We have now integrated comprehensive statistical analyses throughout the manuscript. This includes paired statistical tests with Benjamini-Hochberg correction for all performance comparisons (Figs. R11c, R12, R14, R15) and the use of objective quantitative metrics (LISI, Silhouette Score) to support our UMAP visualizations (Figs. R11a, R13). This addition ensures the statistical robustness of our claims (as detailed in our responses to **Comments 4, 8 & 9**).

We believe these comprehensive revisions significantly enhance the manuscript’s clarity, robustness, and practical applicability. We provide detailed responses to each specific comment below.

Comment 1

In the visual concept generation module, the selection of representative patches requires both GPT-4o filtering and manual validation by human experts. This dependency on expert input may limit the scalability and automation potential of the proposed method, particularly in clinical settings where expert time is costly and throughput demands are high.

Response:

Thank you for raising this important point. We have revised the manuscript to clarify this aspect. The expert-guided selection of representative patches is a one-time offline procedure required only during the initial model training phase. This process is efficient, as it requires expert validation on only a small set of representative patches (10 per class), thereby minimizing the demand on expert time. Once trained, the model is fully automated and highly scalable at inference. It processes new whole-slide images and generates

predictions without needing the original patches or any further human intervention, as illustrated in our inference pipeline (Fig. R10c). The expert knowledge is encoded within the model’s parameters, ensuring high throughput suitable for clinical workflows. Furthermore, this initial investment provides lasting value. When adapting the model to new data, such as fine-tuning on data from similar domains, re-selecting patches is not required, as the initial expert-validated concepts provide a strong, generalizable foundation. In summary, our approach uses a minimal, one-time expert investment to build a robust foundation, which in turn enables a fully automated and scalable model for clinical deployment. In the future, we will develop more automatic strategy to further reduce human intervention. We have added a statement to the Methods section to describe this one-time offline procedure and its implications for scalability.

Comment 2

The use of GPT-4o for generating textual concepts is not accompanied by a validation or correction mechanism. Given potential issues with hallucination and inconsistency in large language models, relying solely on GPT-4o outputs for guiding feature enhancement may introduce potential inaccuracies, especially in high-stakes medical applications.

Response:

Thank you for raising this critical point. Our methodology incorporates a two-layer safeguard to ensure the accuracy and reliability of the textual concepts used to guide FLEX.

First, our methodology includes a **Human-in-the-Loop Expert Validation** step. As an integral part of our workflow, all textual prompts generated by GPT-4o were reviewed and validated by a board-certified pathologist (Fig. R10b). This verification step ensures that all prompts are medically accurate, clinically relevant, and free from factual errors or hallucinations.

Second, we employ a **Constrained and Fact-Based Generation** strategy. The risk of hallucination is inherently minimal because the model is not tasked with creating open-ended narratives but with generating simple descriptive labels grounded in established oncologic terminology (*e.g.*, “invasive ductal carcinoma”, “connective tissue”). This fact-based approach minimizes the scope for LLM-induced inaccuracies, making the subsequent expert validation both efficient and effective.

Furthermore, like the visual concept selection, this concept generation and validation process is a one-time offline procedure that places no burden on the model at inference.

Overall, this two-stage protocol provides a robust defense against potential inaccuracies. We have updated the Methods section to detail this validation process.

Revision in Page 17, Line 935

Textual concept generation. The process of textual concept generation is illustrated in Fig. 6b. It involves creating task-specific and task-agnostic prompts that serve as positive and negative supervision signals. Recognizing the need for accuracy in medical applications, our methodology incorporates an expert validation protocol. First, the prompt generation is constrained to simple, descriptive labels grounded in established oncologic terminology (*e.g.*, “invasive ductal carcinoma”, “connective tissue”), which minimizes the risk of factual inaccuracies. Second, and most critically, we implemented a human-in-the-loop expert validation process. All textual concepts, whether initially derived from established class names or drafted with assistance from GPT-4o, were rigorously reviewed, edited for clinical precision, and formally approved by a board-certified pathologist. This expert oversight ensures that every prompt used is medically accurate and relevant to the prediction task. With these controls, prompts were defined as follows: for tasks with clear class boundaries, such as BRCA subtyping, task-specific prompts are the class names themselves, following previous studies. However, for more complex tasks with vague class boundaries, high complexity, or limited data, class names alone may be insufficient. To address this, we propose using the same class names as task-specific prompts attached to learnable contexts. These contexts adaptively capture task-specific information from the images during training, enhancing the prompts’ effectiveness. Task-agnostic prompts consist of general descriptions irrelevant to the task, such as the names of other tissue types

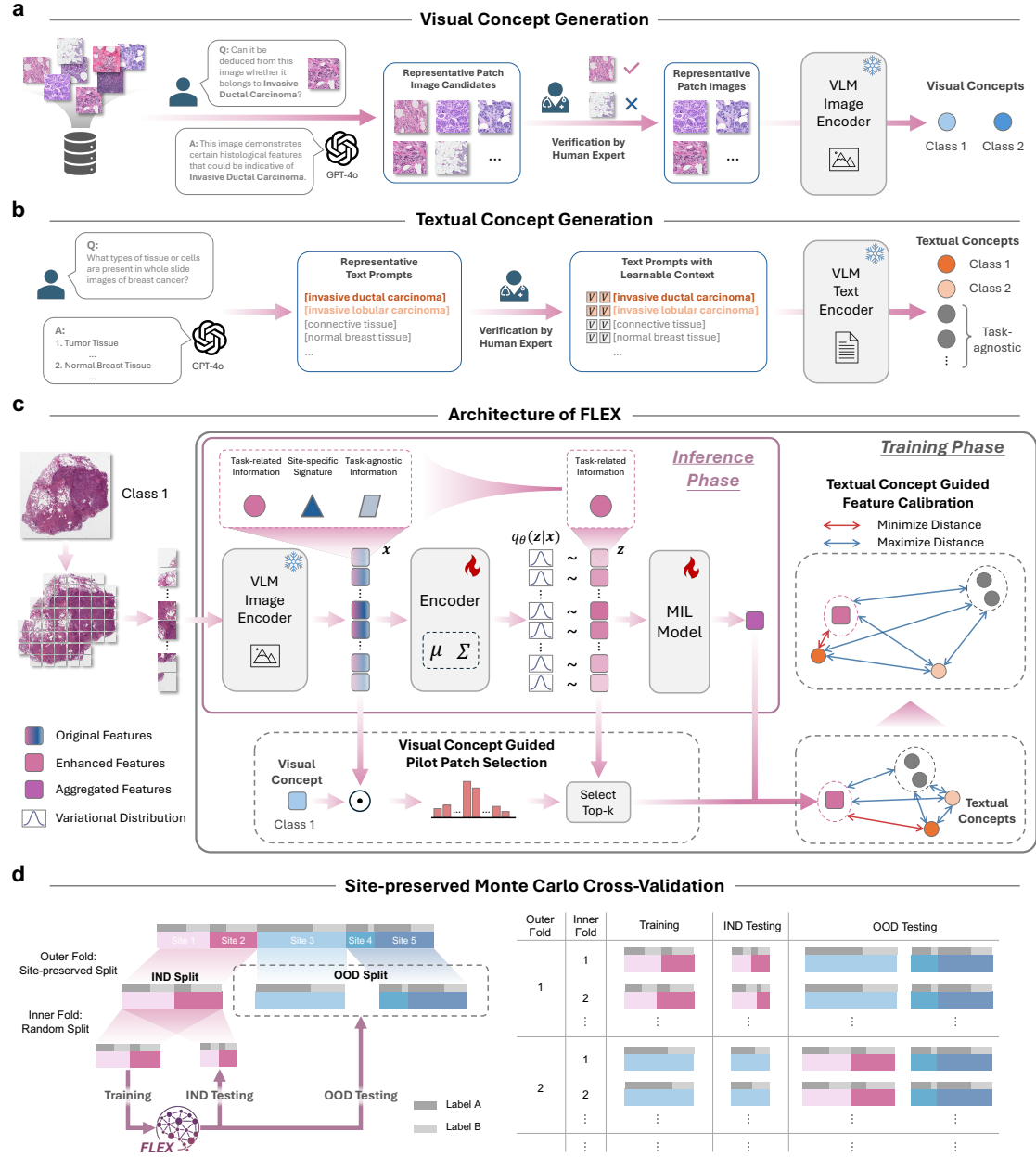


Figure R10: (Figure 6 in manuscript) Overview of the proposed framework and experimental setup. **a**, Visual knowledge generation pipeline. This is a one-time offline procedure during training. ABMIL identifies high-attention patches, followed by GPT-4o-based filtering and human expert verification to select representative patches. Class-specific visual concepts are generated by averaging features from representative patches using the VLM image encoder. **b**, Pipeline for textual knowledge generation, also a one-time offline procedure. GPT-4o is prompted to create representative text prompts (both task-specific and task-agnostic). After expert verification, these prompts are enriched with learnable, data-driven contexts and encoded by the VLM text encoder to create the final textual concepts. **c**, The architecture of FLEX. During inference, original patch features are extracted by the VLM image encoder; a variational encoder then generates parameters for a Gaussian distribution for each patch, from which enhanced features are sampled and aggregated by the MIL model. During training, a Visual Concept Guided Pilot Patch Selection module uses the pre-computed visual concept to select the top-k most relevant enhanced patches. These selected patches are then used in the Textual Concept Guided Feature Calibration process, where an InfoNCE loss aligns the features with the textual concepts by minimizing the distance to the corresponding class concept while maximizing their distance from other concepts. This process helps to optimize the variational encoder of FLEX. **d**, Schematic of the SP-MCCV strategy. The dataset is partitioned into outer folds based on the clinical site to create distinct training and OOD test sets. Inner folds are then used to randomly split the training data for IND evaluation.

(*e.g.*, “connective tissue” or “normal tissue”). These prompts guide the feature enhancement process to filter out task-agnostic information. These are also attached to learnable contexts to improve their effectiveness in filtering out irrelevant features. Finally, textual concepts are generated by averaging the embeddings of these prompts (with their learnable contexts) produced by the VLM’s text encoder.

Comment 3

Although the method is described as a cross-domain generalization method, the actual performance improvements on out-of-domain (OOD) datasets are modest. In some cases, the proposed method (Feature-Level Enhancement for Cross-domain generalization; FLEX) reduces the performance gap from only 4% to 3%, which may have limited clinical relevance or operational impact.

Response:

Thank you for this insightful comment. We will address this from the following two perspectives.

First, while the reduction in the performance gap between IND and OOD settings is one metric, we argue the more critical metric is the absolute performance improvement brought by FLEX on the unseen OOD data. For example, after applying FLEX, the average OOD AUROC was increased by 6.45% (0.853 to 0.908), 7.21% (0.721 to 0.773), and 4.94% (0.628 to 0.659) for morphology classification, molecular biomarker prediction, and gene mutation prediction tasks, respectively, with all results being statistically significant ($p < 0.001$) (Fig. R11c). This represents a substantial and statistically significant improvement on data from unseen clinical sites.

Second, to provide definitive evidence of clinical relevance, we evaluated FLEX on two independent external cohorts: CPTAC and our private in-house NFH dataset. This setup simulates real-world deployment in a new clinical environment. The results (Fig. R11c and Fig. R15) confirm that the performance gains translate directly to these new settings. FLEX achieved significant AUROC increases of 3.5% on CPTAC (0.679 to 0.714, $p < 0.001$) and 3.8% on the in-house NFH dataset (0.733 to 0.771, $p < 0.001$).

These findings demonstrate that the improvements delivered by FLEX are significant and robust, offering tangible operational impact for real-world clinical deployment.

Comment 4

In Figure 2a, the UMAP visualizations show overlapping clusters even after applying the proposed method, indicating that site-specific signatures may not be fully eliminated. A quantitative measure of intra- and inter-class distances or silhouette scores would strengthen the claims of feature disentanglement and clustering quality.

Response:

Thank you for this valuable feedback. Following your recommendation, we have performed a comprehensive quantitative analysis.

To measure the integration of features from different clinical sites, we used the Local Inverse Simpson’s Index (LISI). Our method achieves a LISI score of 4.18, a significant 38% improvement over the original features (3.03) and the Reinhard baseline (3.29). This quantitatively confirms that FLEX substantially mitigates site-specific batch effects.

Crucially, mitigating site effects must not come at the cost of biological signal. To measure class separability, we used the Silhouette Score. Here, FLEX achieves a score of 0.38, a 153% improvement over both the original (0.15) and Reinhard-normalized features (0.14). This demonstrates that as FLEX removes confounding site information, it simultaneously makes the underlying biological classes (IDC vs. ILC) significantly more distinct.

We have revised Figure 2a (now Fig. R11a) to include these quantitative scores and to present the UMAPs colored by both site and biological class separately. This updated figure, backed by LISI and Silhouette metrics, provides clear and robust evidence for our central claim: FLEX successfully disentangles site-specific noise while enhancing the separation of true biological signals.

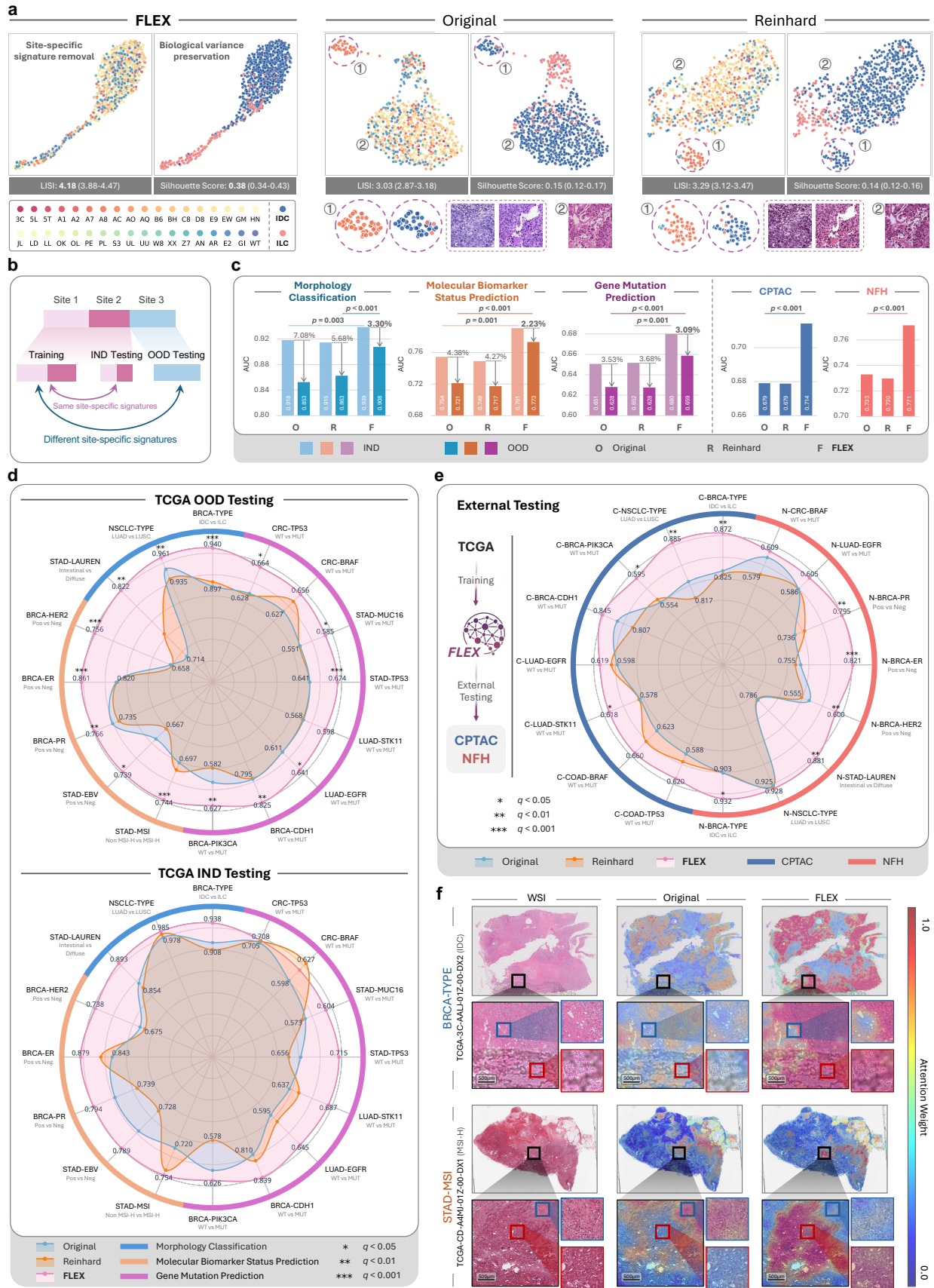


Figure R11: (Figure 2 in manuscript) FLEX enhances model generalizability and diagnostic performance by mitigating site-specific signatures. **a**, UMAP visualization of patch features from the TCGA-BRCA dataset (BRCA-TYPE task), encoded by a CONCH model. Features are colored by submitting site (left panel) or diagnostic class (right panel). Original features and those after Reinhard stain normalization show strong clustering by site and poor separation of diagnostic classes, confirmed by low site integration (LISI: 3.03 and 3.29) and class separability (Silhouette Score: 0.15 and 0.14) scores. Callouts highlight patches from the same site with different diagnoses clustering together, indicating that site-specific signals overpower biological ones. In contrast, FLEX-processed features show effective removal of site signatures (LISI: 4.18) and improved separation of diagnostic classes (Silhouette Score: 0.38). **b**, Schematic of the site-preserved cross-validation setup, defining in-domain (IND) test sets (from sites seen during training) and out-of-domain (OOD) test sets (from unseen sites). **c**, Aggregate AUROC performance across task categories. On both IND and OOD splits of TCGA and on two external cohorts (CPTAC and NFH), FLEX significantly outperforms baselines (paired t-test, $p < 0.01$). **d**, Radar plots of per-task OOD (top) and IND (bottom) AUROC on 16 TCGA tasks. FLEX (pink) consistently outperforms the Original (blue) and Reinhard (orange) methods across nearly all tasks. Statistical significance (paired t-test with Benjamini-Hochberg adjustment; $*q < 0.05$, $**q < 0.01$, $***q < 0.001$) is shown. **e**, Zero-shot generalization performance on external CPTAC and NFH cohorts using models trained only on TCGA. **f**, Attention maps from an ABMIL model for BRCA-TYPE and STAD-MSI tasks. The Original model’s attention is often unfocused or distracted by artifacts. The FLEX-enhanced model correctly localizes its attention to relevant tumor regions, leading to more accurate and interpretable predictions.

Comment 5

Interestingly, standard stain normalization via the Reinhard method achieves better results than their methods in a few tasks. The authors could further discuss reasons leading to this discrepancy and address questions about the added value of their method.

Response:

Thank you for this insightful comment. We believe this observation highlights the distinction between specialized pre-processing and comprehensive feature-level generalization.

The Reinhard method is designed to correct one specific source of variation: stain color. In tasks where stain color is the dominant domain shift, this targeted approach can be effective. However, domain shifts in pathology are rarely this simple; they often involve a complex interplay of variations from tissue preparation, scanner artifacts, and other latent biases. As our UMAP analysis shows (Fig. R11a), complex site-specific feature clusters persist even after Reinhard normalization, demonstrating that stain correction alone is insufficient. While Reinhard performs well in some IND testing scenarios, it does not generalize as well to the broader challenges of OOD testing, where FLEX excels (Fig. R11d).

The added value of FLEX is its more holistic solution, offering advantages in three key areas:

- **Superior Overall Generalization:** FLEX delivers consistently superior performance across the vast majority of tasks, particularly in the more challenging OOD setting (Fig. R11d). Its ability to address the full spectrum of site-specific biases results in stronger and more reliable generalization.
- **Enhanced Fairness:** FLEX addresses critical challenges that simple pre-processing cannot. As shown in Fig. R12, it significantly reduces performance disparities across demographic groups, an essential capability for the equitable deployment of medical AI that is beyond the scope of stain normalization.

In summary, while Reinhard is a valuable tool for a single source of variation, FLEX is a comprehensive framework for domain generalization. Its ability to build more robust, fair, and broadly applicable models provides a clear advantage. We have revised the manuscript to include this discussion.

Comment 6

In Figure 2d and related visualizations, the authors mentioned their methods reduced attention paid to noisy regions. However, what constitutes noise was not clearly defined or objectively measured.

Response:

Thank you for this constructive feedback. We define “noisy regions” as any visual information within a slide that is irrelevant to the specific diagnostic task. This includes two main categories: (1) processing artifacts, such as tissue folds, air bubbles, out-of-focus areas, and other technical imperfections; and (2) task-agnostic biological structures, such as adipose tissue, non-neoplastic stroma, or necrosis when the goal is to classify tumor subtypes. We have now added this definition to the manuscript for clarity.

— Revision in Page 6, Line 323 —

...Here, “noisy regions” are defined as visual information irrelevant to the diagnostic task, such as processing artifacts or task-agnostic biological structures (*e.g.*, non-neoplastic stroma)...

While the gold standard for objective measurement would be to quantify attention against fine-grained regions annotated by expert pathologists, such an effort is beyond the scope of this study. We therefore present the attention maps in Figure 2f as qualitative visualizations that provide an intuitive explanation for our quantitative results. The most objective evidence of our method’s success lies in the significant and consistent improvements in AUROC and fairness metrics across tasks. These robust performance gains serve as definitive proof that FLEX successfully guides the model to focus on task-relevant patterns while suppressing the influence of these distracting signals.

We have also updated our Discussion section to acknowledge this limitation and propose the quantitative validation of attention against pathologist annotations as an important direction for future research.

— Revision in Page 12, Line 651 —

...Future research will focus on expanding FLEX’s versatility. Key directions include broadening compatibility to vision-only foundation models and extending its application to a wider range of tasks, including regression and prognosis. We will also explore its integration with pathology vision large language models and work to enhance the interpretability and adaptability of these models. This effort includes the quantitative validation of attention mechanisms against expert-annotated regions by pathologists, which would provide an objective confirmation of the model’s focus. Such work will be crucial for translating these advancements into clinically impactful and trustworthy tools.

Comment 7

In Figure 3, the fairness gap (measured in AUROC differences across racial groups) ranges widely, sometimes as high as 0.7. Surprisingly, the proposed methods exhibit the widest performance spread in certain tasks. The authors could discuss the implications of this observation and the potential impact on the reliability of the proposed methods.

Response:

Thank you for this crucial observation. The wide performance spread in certain tasks is a statistical artifact of confidence interval calculation, which can be highly sensitive to single outliers across the 15 validation folds, rather than an indicator of unreliability. We have provided more detailed analysis to demonstrate that FLEX is consistently more reliable at reducing fairness gaps.

As shown in the granular scatter plots (Fig. R12a), even in tasks like the CRC-BRAF where the error bar is wide, the effect is often driven by a single outlier, while the main cluster of FLEX results still shows superior performance.

More importantly, a comprehensive assessment requires looking beyond a single metric. Our analysis of True Positive Rate (TPR) Disparity (Fig. R12c) offers a more direct measure of equitable performance. The violin plots for FLEX are visibly and consistently narrower and more tightly centered around the zero-disparity line across nearly all tasks. This provides strong evidence that our method enables the model to perform more reliably across different demographic groups. This conclusion is further substantiated by the quantitative TPR Disparity RMSE in Fig. R12d, where FLEX achieves a significantly lower (*i.e.*, better) disparity score. In summary, while the confidence interval for one metric can be skewed, the totality of the evidence demonstrates that FLEX consistently improves both fairness and model reliability. We have revised our

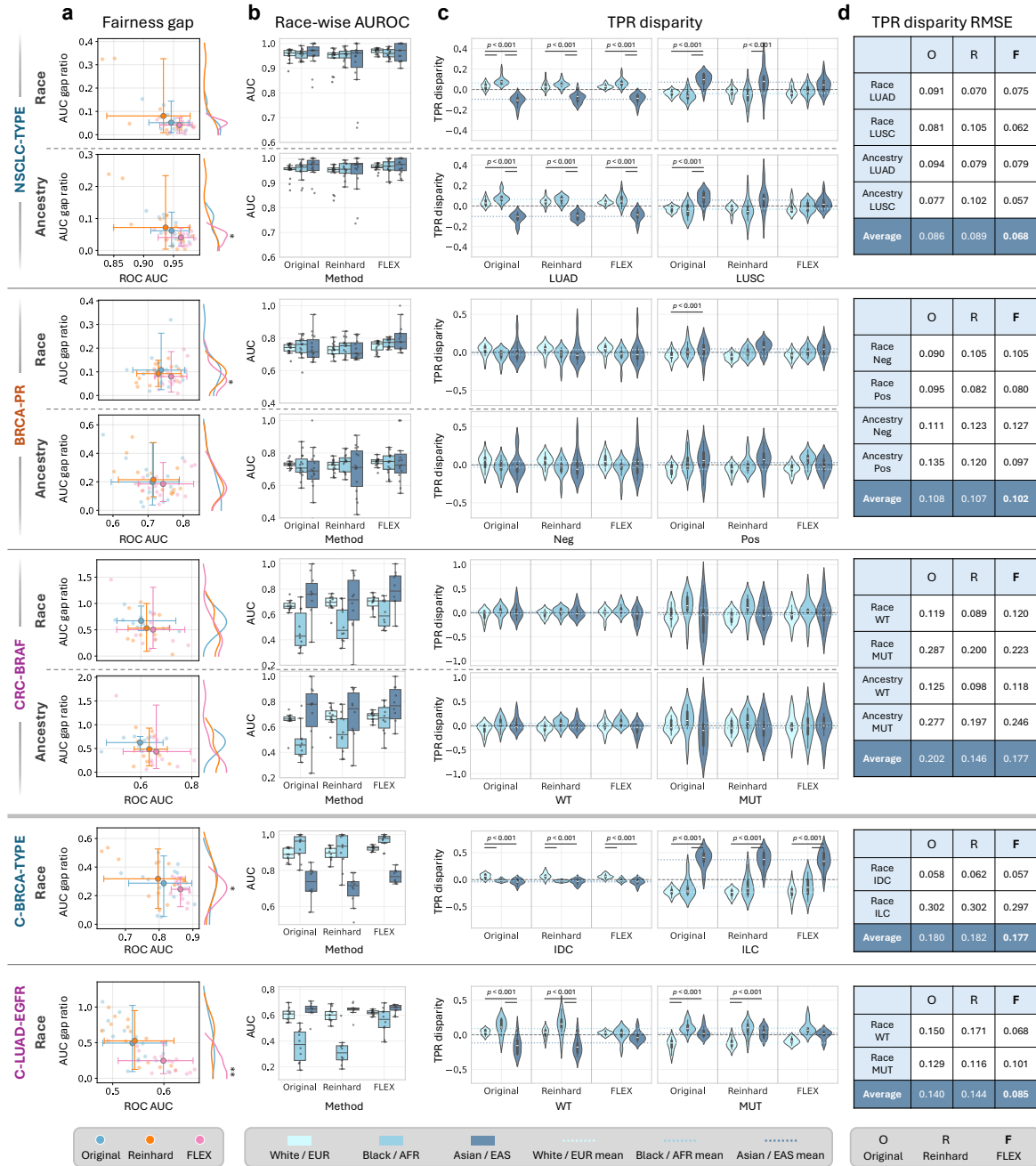


Figure R12: (Figure 3 in manuscript) Evaluation of FLEX's effectiveness in improving demographic fairness. Fairness is evaluated across race and ancestry groups using results from 15 cross-validation folds. For each fold, metrics for each demographic-label group were estimated via bootstrapping (500 resamples). **a**, The fairness gap for five representative tasks. The AUC gap ratio is the performance difference between the best- and worst-performing subgroups relative to the overall AUC. Smaller dots represent individual folds; large dots with error bars show the mean and 95% CI. A lower ratio indicates better fairness. Marginal distributions show the distribution of AUC gap ratios across folds. Statistical significance ($*p < 0.05$, $**p < 0.01$) from a paired Wilcoxon signed-rank test denotes the improvement of FLEX over the Original method. **b**, The AUROC distribution across subgroups. Boxes indicate the interquartile range of the AUROC distribution ($n = 15$ folds), with the center being the 50th percentile and the whiskers extending to data points within $1.5 \times$ the interquartile range. Tighter distributions indicate more equitable performance. **c**, Violin plots of True Positive Rate (TPR) disparity, defined as the difference between a subgroup's TPR and the overall TPR. Distributions centered closer to the zero-disparity line indicate higher fairness. P-values from the paired t-test are shown. The dashed line represents the overall TPR across three methods. **d**, Quantitative summary of TPR disparity, showing the Root Mean Square Error (RMSE) for Original (O), Reinhard (R), and FLEX (F). Lower values indicate better fairness.

manuscript to guide the reader through this multi-faceted interpretation to avoid any misimpression and to clarify the robust reliability of our method.

Comment 8

While the proposed method is compared against baseline and the standard Reinhard stain normalization method in Figure 3, no statistical tests (*e.g.*, paired t-tests or Wilcoxon signed-rank tests) are reported to determine whether the observed performance differences are significant. The observed differences in results between methods appear minimal at times.

Response:

Thank you for this crucial suggestion. In our revised manuscript, we have performed a comprehensive statistical analysis, with results presented in the updated Figure 3 (Fig. R12). These tests underscore the significance of our method’s improvements.

For the AUC Gap Ratio, we applied the non-parametric paired Wilcoxon signed-rank test. As indicated by the newly added significance markers in Fig. R12a, FLEX achieves a statistically significant reduction in the fairness gap in multiple key tasks, including NSCLC-TYPE and BRCA-PR. This finding also holds for the external CPTAC dataset (*e.g.*, C-BRCA-TYPE and C-LUAD-EGFR), confirming that the fairness improvements delivered by FLEX are robust and generalizable.

For the TPR Disparity, paired t-tests similarly confirm that FLEX reduces disparity in most tasks, with eliminated significant differences between different groups on several individual tasks, such as NSCLC-TYPE and BRCA-PR.

Furthermore, we have added comprehensive statistical tests to other analyses, such as the AUROC distributions for each task (Fig. R11c, d, e) and performance gains across different MIL model variants (Fig. R14c). These analyses provide a solid foundation for the claims made in our manuscript, demonstrating the robustness of FLEX’s improvements.

Comment 9

In Figure 4b, UMAP plots for the features obtained from the proposed methods still show substantial overlap between class labels across sites. The visual evidence does not compellingly support the claim of improved separability of diagnostic categories.

Response:

Thank you for this valuable feedback. To provide a more objective evaluation, we have augmented Figure 4b (now Fig. R13) with quantitative metrics that directly measure FLEX’s two key objectives: harmonizing features across sites and improving the separability of diagnostic classes.

First, to quantify the removal of site-specific batch effects, we calculated the Local Inverse Simpson’s Index (LISI). The results demonstrate that FLEX increases the LISI score by 68% for PathGen-CLIP (from 1.75 to 2.94) and 45% for CONCH (from 2.22 to 3.22), providing quantitative proof of a significantly more harmonized feature space.

Second, to objectively measure class separability, we used the Silhouette Score. The score for CONCH more than doubles from 0.09 to 0.24, and other models also show consistent positive gains.

While some visual overlap is expected in such challenging classification tasks, the LISI and Silhouette scores demonstrate that FLEX systematically refines the underlying feature geometry to achieve both better site integration and enhanced class separability. We have updated the manuscript and figure caption to include this analysis, making our conclusions more objectively verifiable.

Comment 10

In Figure 5, performance improvements across different data scales and MIL model variants are relatively small, sometimes between 0.000 and 0.005 in AUROC. The authors could discuss the practical benefits of integrating the proposed methods into real-world pathology diagnosis.

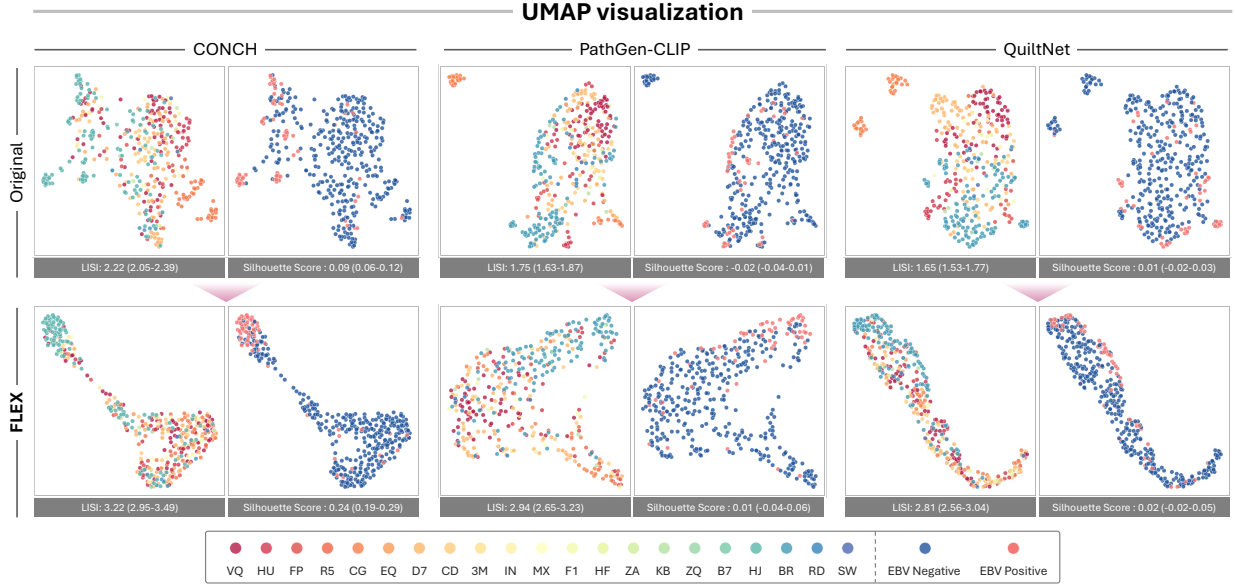


Figure R13: (Figure 4b in manuscript) UMAP visualizations illustrating the effect of FLEX on the patch feature space for the STAD EBV task. For each VLM, parallel subplots are colored by site (left) to visualize batch effect mitigation, and by EBV status (right) to visualize class separability. The Local Inverse Simpson’s Index (LSI) for site integration (higher is better) and the Silhouette Score for class separation (higher is better) provide quantitative evidence. The visualizations and scores demonstrate that FLEX reduces site-specific clustering while improving the discriminability of task-relevant classes.

Response:

Thank you for this important point. We have strengthened our manuscript to address the practical benefits of FLEX through two key lines of evidence: statistical robustness and superior generalization to unseen clinical data.

First, we established the statistical significance of these gains First, we established the statistical significance of these gains within the TCGA dataset. Our learning curve analysis (Fig. R14b) shows that after correcting for multiple comparisons, the improvements from FLEX are statistically significant across all tested data scales ($q < 0.05$), proving the gains are reliable. This benefit is also universal across architectures. As visualized in Fig. R14c, FLEX consistently and significantly improves performance across five different MIL models, with a highly significant average improvement for every model ($q < 0.001$). This confirms FLEX acts as a robust and broadly beneficial module.

However, the most crucial measure of practical benefit lies in a model’s ability to generalize to new, real-world data. To this end, we conducted extensive zero-shot validation on two independent external cohorts: the public CPTAC dataset ($n=1,738$ slides) and our in-house NFH dataset ($n=5,318$ slides). The results (Fig. R15) demonstrate that models trained with FLEX on TCGA data show a markedly superior ability to generalize. The average AUROC across all tasks significantly improved on both external cohorts ($p < 0.001$, Fig. R15a), and task-specific results show consistent outperformance (Fig. R15b,c, and d). This result is critical, as it demonstrates that FLEX effectively mitigates dataset-specific biases, addressing a primary hurdle for deploying AI in real-world diagnostic settings.

In summary, the practical benefit of FLEX is twofold: it delivers a performance boost that is consistent and statistically significant, and more importantly, it enhances generalization to unseen clinical data. We have incorporated this discussion into the manuscript to highlight the practical relevance of our method.

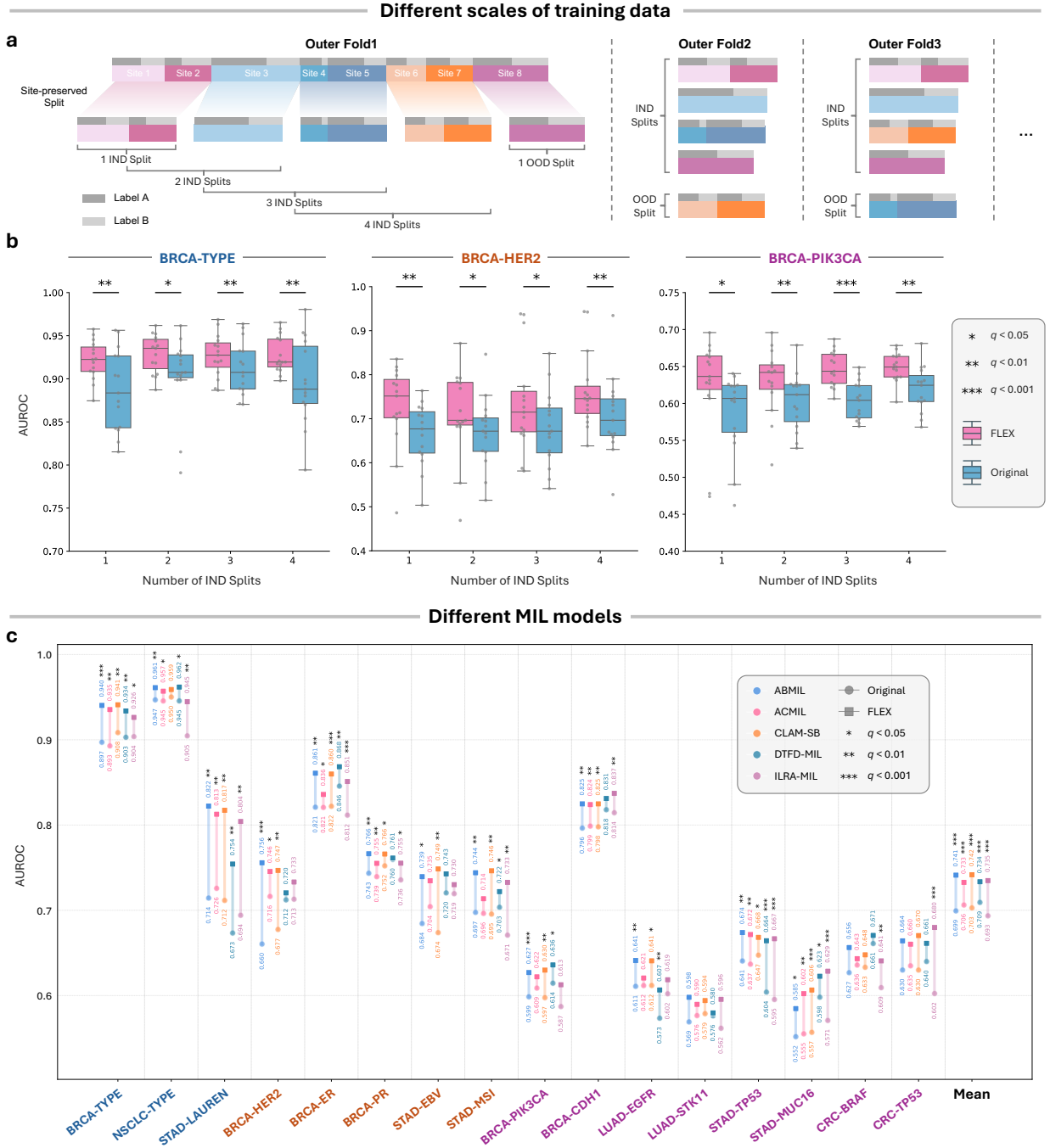


Figure R14: (Figure 5 in manuscript) Performance of FLEX across training data scales and MIL models. **a**, Schematic of the experimental design for evaluating performance at different training data scales. The procedure follows our main Site-Preserved Monte Carlo Cross-Validation (SP-MCCV) strategy, which involves 5 outer folds and 3 inner Monte Carlo Cross-Validation (MCCV) folds, for a total of 15 evaluation runs. Specifically for this experiment, the dataset is first partitioned into 5 site-preserved outer folds. In each of the 5 outer loop iterations, one fold is held out as the OOD test set. Training sets of increasing size (1, 2, 3, and 4 folds) are constructed from the remaining 4 folds to create a learning curve. For each scale condition, 3 inner Monte Carlo Cross-Validation (MCCV) runs are performed to ensure stable results, for a total of 15 evaluation runs per condition. **b**, OOD AUROC comparison between FLEX and baseline (Original). Box plots show aggregated results from 15 runs for each training data scale. Asterisks indicate statistically significant improvement (Benjamini-Hochberg adjusted q -value: $* < 0.05$, $** < 0.01$, $*** < 0.001$). Box plots show Q1–Q3 interval with the median; whiskers are $1.5 \times$ interquartile range. **c**, Dumbbell plot showing OOD AUROC after integrating FLEX with five state-of-the-art MIL models across 16 tasks. FLEX consistently improves performance across different MIL architectures.

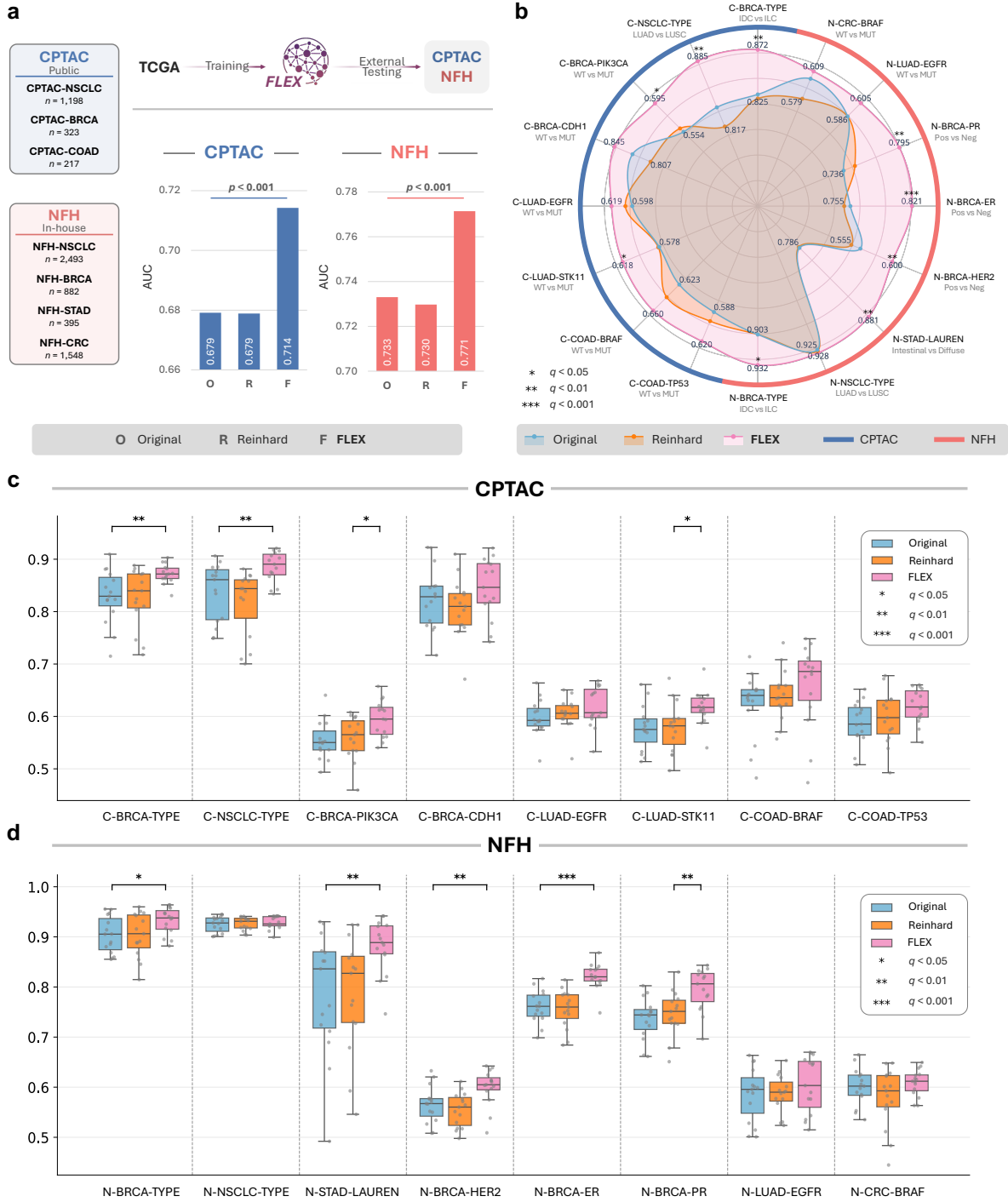


Figure R15: Zero-shot generalization of TCGA-trained models to external CPTAC and in-house NFH cohorts. Models were trained exclusively on TCGA data and evaluated directly on two independent external datasets without fine-tuning. **a**, Overall experimental design and results. Bar plots display the average AUROC across all tested tasks on the CPTAC (left) and in-house NFH (right) datasets. FLEX significantly improves the average performance of the TCGA-trained model on both external cohorts ($p < 0.001$, paired t-test). “O”, “R”, and “F” denote Original, Reinhard, and FLEX, respectively. **b**, Radar plot comparing task-specific AUROC across models on CPTAC and NFH data. FLEX consistently expands the performance radius across tasks. Asterisks indicate statistical significance of the improvement by FLEX compared to the original model (* $q < 0.05$, ** $q < 0.01$, *** $q < 0.001$, Benjamini-Hochberg FDR correction). **c**, **d**, Detailed box plots showing task-by-task performance on the (c) CPTAC and (d) NFH cohorts. FLEX consistently outperforms or matches the baseline and Reinhard-normalized models, demonstrating its robust ability to enhance zero-shot transfer learning by mitigating dataset-specific biases. Boxes indicate quartile values of the AUROC distribution ($n = 15$ folds), with the center line indicating the median and whiskers extending to $1.5 \times$ the interquartile range.

Responses to Reviewers' Comments for Manuscript NCOMMS-25-16314A

**Knowledge-Guided Adaptation of Pathology
Foundation Models Effectively Improves Cross-domain
Generalization and Demographic Fairness**

Addressed Comments for Publication to
Nature Communications

Authors' Response to Reviewer 2

Comment 1

Remarks to the Author: The additional analysis of two large datasets and subsequent results and manuscript updates largely address my concerns.

Response:

Thank you very much for your positive feedback and for acknowledging that our revisions have successfully addressed your concerns.

Comment 2

Remarks on code availability: The code for the initial submission is included but code relating to resubmission updates is missing from the repository. The code is well organized and the repository has a README with instructions to run the code.

Response:

Thank you for your careful examination of our code repository and for your positive comments on its organization. To make the codebase clearer for future users and to ensure full reproducibility, we have updated the README.md file in our repository. This updated documentation now explicitly details the steps and parameters required to replicate the results using existing code.

Authors' Response to Reviewer 3

Comment 1

Remarks to the Author: The authors have addressed the previously raised comments. Thank you.

Response:

We sincerely appreciate your confirmation and are happy to know that our revisions have met your expectations.

Comment 2

Remarks on code availability: The authors provided a README file with instructions for installing and running the application.

Response:

We appreciate your positive comment on the usability and documentation of our code.